

Fluctuations of the type D ASEP: decoupling in two universality classes

(working draft — full version)

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Abstract

The type D ASEP is a two-species interacting particle system on $\mathbb{Z}$, the $n \rightarrow \infty$ degeneration of the $U_q(\mathfrak{so}_{2n+2})$ exclusion process of Kuan, Landry, Lin, Park and Zhou, in which two separately conserved species hop, bind into a composite “bound pair”, and split. The binding interaction couples the two species at the microscopic level as directly as a conservative dynamics can; the question we ask is whether that coupling survives to the scale of the fluctuations. We analyse the density and current fluctuations in two scaling regimes and prove that in both the two species *decouple* — but for two different reasons.

In the critically weak-asymmetry (Edwards–Wilkinson) regime $q = 1 - c/N^2$ we prove that the two density fluctuation fields *decouple*: each converges to a linear stochastic heat equation, with no cross-coupling in either the drift or the noise, the limiting noises having vanishing cross-correlation (Theorem 6.1). The proof is self-contained modulo the classical single-species fluctuation result: the drift reduces, via the exact current decoupling (Proposition 3.1), to an autonomous single-species WASEP, while the cross-noise cancellation is controlled by a two-particle dual kernel bound (Theorem 5.2) proved here in full, including the underlying one-dimensional local limit estimates (§4). The inter-species coupling the model nonetheless exhibits is shown to be purely an initial-condition effect, computed in closed form through the two-particle dual (Theorem 7.5): a position correlation $(1 - e^{-4c})/(4c)$ and a Bessel–Struve form for the q -Krawtchouk observable, both with $1/c$ tails.

In the fixed- q (Kardar–Parisi–Zhang) regime an exact current–decoupling identity, valid at every n (Propositions 3.1, 3.2), makes each species’ current an autonomous single-species ASEP, hence Tracy–Widom for every n — proving a conjecture of [3]. Here too the species decouple: we prove an exact contact representation for the joint q -Laplace observable — $\text{Cov}(q^{2N_1}, q^{2N_2})$ is a weighted sum of connected two-particle dual hitting probabilities, vanishing identically without the binding (Theorem 8.4) — while for the linear currents simulations give $\text{Cov}(N_1, N_2) = (0.099 \pm 0.003)\sqrt{s}$ at $q = \frac{1}{2}$, so that $\text{Corr}(N_1, N_2) \asymp s^{-1/6} \rightarrow 0$ (Conjecture 8.5). The common engine is the q -Krawtchouk orthogonal self-duality, together with the fact — which we make precise — that the inter-species coupling is invisible to every equilibrium transport coefficient, so that in both regimes what remains of it is a transient, initial-condition-driven correlation rather than a feature of the limiting law.

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1 Introduction

When several conserved species share one exclusion dynamics, the basic question is whether they interact at the scale of their fluctuations: do the density and current fields couple in the scaling limit, or does each species behave as an autonomous single-species process? Coupling is the generic expectation, and it is what produces the rich behaviour of multi-species models — off-diagonal transport coefficients, distinct characteristic speeds, and the non-Gaussian or non-Tracy–Widom joint laws that come with them. This paper studies a model in which the answer runs the other way. The two species are not merely placed on a common lattice: they are dynamically *fused* into composite pairs and later *split*, so any inter-species coupling has a direct microscopic channel through which to act. Yet in two distinct scaling limits the two components decouple completely, and our aim is to locate exactly where the interaction goes — and why it never reaches the limiting law.

1.1 The model

Fix $q \in (0, 1)$. The *type D ASEP* was introduced by Kuan, Landry, Lin, Park and Zhou [12], who constructed it from the quantum groups $U_q(\mathfrak{so}_6)$ and $U_q(\mathfrak{so}_8)$ ($n = 2, 3$ in $\mathfrak{so}_{2n+2}$, Cartan type D); the construction has been extended to $U_q(\mathfrak{so}_{10})$ [17], and the explicit rate matrix (displayed for general n in [3, §2.1]) makes sense for every $n \in \mathbb{N}$. It is a continuous-time Markov process of two conserved species on $\mathbb{Z}$ which, beyond hopping, may bind into a composite *bound pair* and later split. This binding is what makes the model a stringent test of decoupling: the two species are genuinely interacting, not two independent copies sharing the exclusion rule. Blyschak, Burke, Kuan, Li, Ustilovsky and Zhou [3] established an orthogonal q -Krawtchouk self-duality for $n = 2, 3$

and conjectured it for all n ; we verify it at the $n = \infty$ degeneration used here (Theorem 2.6(ii)). The duality is the analytic engine of this paper: it reduces every fluctuation question to a computation with a handful of dual particles.

We work in the limit $n \rightarrow \infty$, in which (after rescaling time by q^{2n}) the dynamics closes on the four local states $\{0, 1, 2, 3\}$ at each site: empty, species 1, species 2, and a bound pair carrying one particle of each species. With $\eta_{i,x} \in \{0, 1\}$ the occupation of species i at x , so that $\omega_x = \eta_{1,x} + 2\eta_{2,x}$, and with $\varepsilon := 1 - q^2$, the nonzero nearest-neighbour rates are

$$\begin{aligned} (1, 0) \rightarrow (0, 1) : 1, \quad (0, 1) \rightarrow (1, 0) : q^2, \quad (2, 0) \rightarrow (0, 2) : 1, \quad (0, 2) \rightarrow (2, 0) : q^2, \\ (3, 0) \rightarrow (0, 3) : 1, \quad (0, 3) \rightarrow (3, 0) : q^4, \quad (0, 3) \rightarrow (1, 2) : q^2\varepsilon, \quad (0, 3) \rightarrow (2, 1) : q^2\varepsilon, \\ (1, 2) \rightarrow (0, 3) : \varepsilon, \quad (1, 2) \rightarrow (2, 1) : q^2, \quad (2, 1) \rightarrow (0, 3) : \varepsilon, \quad (2, 1) \rightarrow (1, 2) : q^2, \end{aligned} \quad (1)$$

together with mixed hops $(3, 1) \leftrightarrow (1, 3)$ and $(3, 2) \leftrightarrow (2, 3)$ at rate 1 right, q^2 left, in which a single species slides past a bound pair. The two species numbers are conserved. For finite n a single species is a usual ASEP with rates $q^{\pm 1}\beta_n$, $\beta_n = q^{1-2n} + q^{2n-1}$, so n enters the single-species dynamics only as a time rescaling.

1.2 Results

Decoupling might hold for one reason in one universality class and fail in another, so we test it in the two canonical fluctuation classes of driven diffusive systems. We treat two scalings. (A) *The Edwards–Wilkinson regime*: asymmetry tuned to the diffusive rate $q = q_N = 1 - c/N^2$, under diffusive space–time scaling, at any pair of densities (ρ_1, ρ_2) . (B) *The Kardar–Parisi–Zhang regime*: $q \in (0, 1)$ fixed, the integrated currents $N_1(s), N_2(s)$ across the origin from the step initial condition, at a single time s . Our principal theorems are Theorems 6.1, 7.5 (regime A) and 8.1, 8.4 (regime B), proved in §§6–8. The logical backbone common to both regimes — the reversible measure and orthogonal duality (§2), the exact current decoupling and vanishing transport coefficients (§3), and a local CLT for the dual coordinates (§4) — is developed first.

1.3 Relation to previous work

Regime (A) belongs to the nonlinear fluctuation theory of weakly asymmetric systems and the passage to stochastic Burgers/KPZ, due to Gonçalves and Jara [9], with uniqueness by Gubinelli and Perkowski and spectral-gap-free refinements by Gonçalves, Jara and Simon [10]. The few-body estimates we need we obtain through the *orthogonal polynomial duality* method of Ayala, Carinci and Redig [1, 2], which reduces them to a decay bound on the few-particle dual kernel. In our setting the exact current decoupling (Proposition 3.1) makes the drift an autonomous single-species WASEP, so the duality enters only for the inter-species cross-noise, controlled by the type D two-particle kernel bound of §5; the would-be slow internal mode of the bound pair never couples to the species current at all (Proposition 6.9).

For regime (B), the single-species Tracy–Widom asymptotics are Borodin–Corwin–Sasamoto [4]; the exact q -Laplace decoupling identity (Theorem 8.4) is to our knowledge new, and the Tracy–Widom marginals at every n (Theorem 8.1) prove the conjecture of [3, §3.1.1].

The structural point is what distinguishes type D from other multi-species models. Its inter-species coupling is invisible to both the static compressibility and the dynamic mobility, so what remains of it is a transient, initial-condition-driven correlation rather than a transport effect. This sets type D apart from the multi-species models with genuine off-diagonal transport coefficients (e.g. multi-species stirring, with multinomial mobility) and from the distinct-velocity two-species

KPZ decouplings: type D sits instead at the *umbilic point*, where the characteristic velocities coincide and both of those mechanisms are unavailable. We are not aware of another model combining diffusive (EW) class, diagonal compressibility, vanishing cross-mobility, and a nonzero shared-source inter-species correlation with a $\rho(c) \sim 1/c$ crossover.

1.4 Organisation

Section 2 fixes the generator, reversible measure, and orthogonal self-duality. Section 3 proves the exact current decoupling and the vanishing static/dynamic cross coefficients. Section 4 proves the random-walk local CLTs used throughout. Section 5 proves the two-particle dual kernel bound. Section 6 proves the decoupled Edwards–Wilkinson limit. Section 7 proves the initial-condition crossover. Section 8 proves the Tracy–Widom marginals at every n and the exact q -Laplace decoupling of the two currents, and states the linear covariance asymptotics as a conjecture. Section 9 records the asymmetry-exponent phase picture and further numerical observations. Section 11 discusses open problems. Physical interpretation is confined to remarks.

2 The model, the measure, and the duality

We first record the three structures on which the entire analysis rests: the generator, the reversible product measure, and the orthogonal q -Krawtchouk self-duality. The duality is the decisive one — it is what turns a question about the full interacting field into a question about one or two dual particles, and both regimes are ultimately solved by the same few-body computation.

For a local function f of $\eta \in \{0, 1, 2, 3\}^{\mathbb{Z}}$ the generator is

$$\mathcal{L}f(\eta) = \sum_{x \in \mathbb{Z}} \sum_{(a,b) \rightarrow (a',b')} r((a,b) \rightarrow (a',b')) \mathbf{1}\{\omega_x = a, \omega_{x+1} = b\} [f(\eta_{a'b'}^{x,x+1}) - f(\eta)], \quad (2)$$

the inner sum over the transitions of (1), $\eta_{a'b'}^{x,x+1}$ denoting η with (ω_x, ω_{x+1}) reset to (a', b') . The finite- n rate matrix is displayed in [3, §2.1] (constructed from the quantum group for $n = 2, 3$ in [12]); after the time rescaling by q^{2n} its entrywise $n \rightarrow \infty$ limit is (1), which one checks directly to be a Markov generator with the stated conservation laws for every $q \in (0, 1)$.

Proposition 2.1 (reversible measure; [12, Prop. 1.3], [3, §3.1]). *The type D ASEP is reversible with respect to the product blocking measure*

$$\nu(\eta) = \mu_{\alpha_1}(\eta_1) \mu_{\alpha_2}(\eta_2), \quad \mu_{\alpha}(\xi) = \prod_{x \in \mathbb{Z}} \alpha^{\xi_x} q^{-2x\xi_x} \quad (\alpha > 0), \quad (3)$$

a product over both sites and species, carrying the q^{-2x} density tilt, the two species independent.

Lemma 2.2 (local detailed balance). *For ν and adjacent sites $(x, x+1)$, the unnormalised two-point weights satisfy*

$$\nu(0, 3) = q^{-4} \nu(3, 0), \quad \nu(1, 2) = \nu(2, 1) = q^{-2} \nu(3, 0). \quad (4)$$

Proof. By (3) the weight on $\{x, x+1\}$ factorises over the two sites and two species; a species- i particle at site y contributes $\alpha_i q^{-2y}$. Moving the bound pair from x to $x+1$ multiplies the weight by $q^{-2} \cdot q^{-2} = q^{-4}$; moving one species-1 particle from x to $x+1$ and one species-2 particle from $x+1$ to x leaves a single q^{-2} deficit. These are precisely the detailed-balance identities of (1) for ν . $\square$

Remark 2.3 (parameter range and the locked limit). Although n indexes finite-dimensional representations of $\mathfrak{so}_{2n+2}$, the finite- n generator of [3, §2.1] extends analytically in n and remains a bona fide Markov generator for a range of *negative* n : the binding rates $(1, 2) \rightarrow (3, 0)$, $(3, 0) \rightarrow (1, 2)$ are the constraints, both proportional to $2 - q^{2n-2}(1 - q^2)$ and hence vanishing at $n_{\text{crit}} = \log(2q^2/(1 - q^2))/(2 \log q)$, so the process is stochastic for $n > n_{\text{crit}}$, with $n_{\text{crit}} < 0$ exactly when $q > 1/\sqrt{3}$ (i.e. $2q^2/(1 - q^2) > 1$). At $q = 1$ the split/merge rates vanish, bound pairs never break, $N_1 \equiv N_2$, and the species are perfectly locked.

2.1 The orthogonal q -Krawtchouk self-duality

Each site holds at most one particle of each species, so the single-site, single-species marginal is the two-point specialisation of the q -Krawtchouk family. With local fugacity $t_x := \alpha q^{-2x}$ and density $\rho_x = t_x/(1 + t_x)$, the degree- ≤ 1 orthogonal polynomials of μ_α are

$$\kappa_0(\eta_x) \equiv 1, \quad \kappa_1(\eta_x) = \frac{(1 + t_x)\eta_x - t_x}{\sqrt{t_x}} = \frac{\eta_x - \rho_x}{\sqrt{\rho_x(1 - \rho_x)}}. \quad (5)$$

Definition 2.4 (orthonormal local basis). For a dual configuration ξ specifying which (site, species) pairs carry a dual particle, set

$$D^{\text{loc}}(\xi, \eta) = \prod_{(x,i): \xi_{i,x}=1} \kappa_1(\eta_{i,x}), \quad (6)$$

a polynomial in η of degree $|\xi|$ (the dual mass). By the product structure of ν and the single-site orthonormality of (5), the family $\{D^{\text{loc}}(\xi, \cdot)\}$ is orthonormal in $L^2(\nu)$. It is the natural basis for the fluctuation fields: mass-one elements are the centred density fields, and the different-species mass-two element at one site is the bound-pair mode.

Remark 2.5 (D^{loc} is *not* the duality function). An earlier version of this draft asserted self-duality with respect to (6). That is false, for a structural reason: every $\int D^{\text{loc}}(\xi, \cdot)^2 d\nu = 1$, and reversibility makes $\langle P_t D_\xi^{\text{loc}}, D_{\xi'}^{\text{loc}} \rangle_\nu$ a symmetric matrix, so duality with respect to (6) would force a symmetric dual transition kernel — impossible for the asymmetric, sticky-origin dual particle dynamics. (We also verified the failure directly: on two and three sites the interlacing $\mathcal{L} D^{\text{loc}} = D^{\text{loc}} \mathcal{L}^\top$ fails by $O(1)$ margins at generic parameters.) The true duality function of [3] carries nonlocal counting factors, as follows.

Theorem 2.6 (orthogonal q -Krawtchouk self-duality). *For configurations η, ξ and $i \in \{1, 2\}$ let $N_{x-1}^-(\xi_i)$ be the number of species- i dual particles strictly to the left of x and $N_{x+1}^+(\eta_i)$ the number of species- i particles of η strictly to the right of x , and set*

$$D_{\alpha_1, \alpha_2}(\xi, \eta) = \prod_{i=1,2} \prod_{x: \xi_{i,x}=\eta_{i,x}=1} \left(1 - \frac{q^{2(x - N_{x-1}^-(\xi_i) + N_{x+1}^+(\eta_i))}}{\alpha_i} \right), \quad (7)$$

the two-point specialisation of the q -Krawtchouk product of [3, Thm. 3.1]. (i) [3, Thm. 3.1]: for parameters $(q, n, \delta) = (q, 2, 0)$ and $(q, 3, 0)$, the type D ASEP is self-dual with respect to (7). (ii) At $n = \infty$, i.e. for the rates (1): the two-site interlacing $\mathcal{L} D = D \mathcal{L}^\top$ holds identically in (q, α_1, α_2) — a finite rational identity, which we have verified by computer algebra — and the induction in the proof of [3, Thm. 3.1, eqs. (16)–(19)], which uses only the two-site case and the fugacity-shift covariance of (7), extends it to any number of sites. This settles the $n = \infty$ degeneration of the conjecture of [3, Rmk. 4] that the duality holds for all n . In the lineage of orthogonal stochastic dualities from U_q -representations [11, 7], this is the type D instance.

Corollary 2.7 (triangular degeneration). *Self-duality with respect to (7) holds identically in (α_1, α_2) , hence coefficient-wise in $\alpha_1^{-1}, \alpha_2^{-1}$; and since the dual generator preserves the particle numbers $(|\xi_1|, |\xi_2|)$, the coefficients of $\alpha_1^{-|\xi_1|} \alpha_2^{-|\xi_2|}$, glued across the dual sectors, again form a duality matrix. This extraction yields self-duality with respect to the triangular function*

$$D^{\text{tri}}(\xi, \eta) = \mathbf{1}\{\xi \subseteq \eta\} \prod_{i=1,2} \prod_{x: \xi_{i,x}=1} q^{2(x - N_{x-1}^-(\xi_i) + N_{x+1}^+(\eta_i))}, \quad (8)$$

the type D analogue of the Schütz and Borodin–Corwin–Sasamoto q -exponential dualities [18, 4]. On configurations η with finitely many particles to the right — in particular on the step initial condition and its evolution — the factors of (8) are uniformly bounded for each fixed dual mass, and (8) is the form of the duality used in §8.

Lemma 2.8 (duality covariance; [1, Lem. 2.1]). *Let $\{D(\xi, \cdot)\}$ be a family of duality functions for the process which is orthogonal in $L^2(\varpi)$ for some measure ϖ , $\int D(\xi, \eta) D(\xi', \eta) d\varpi = \delta_{\xi, \xi'} a(\xi)$. Then for all ξ, ξ' and $t \geq 0$,*

$$\int \mathbb{E}_\eta[D(\xi, \eta_t)] D(\xi', \eta) d\varpi(\eta) = p_t(\xi, \xi') a(\xi').$$

Proof. Insert the duality identity $\mathbb{E}_\eta[D(\xi, \eta_t)] = \sum_{\xi''} p_t(\xi, \xi'') D(\xi'', \eta)$, multiply by $D(\xi', \eta)$, and integrate against ϖ . By orthogonality $\int D(\xi'', \eta) D(\xi', \eta) d\varpi = \delta_{\xi'', \xi'} a(\xi')$, so only $\xi'' = \xi'$ survives. The interchange of sum and integral is justified by the finite mass of ξ (only finitely many ξ'' are reachable in the particle-conserving dual). $\square$

Remark 2.9 (the exact orthogonality relation). The precise orthogonality satisfied by the type D duality functions is the two-point, two-species specialisation of [5, Thm. 3.2], the single-species input cited in [3, §3.1]. On a finite lattice Λ of $|\Lambda|$ sites define the *sector-reweighted* blocking measure and norm functions ($N = N(\zeta)$ the particle number, all up to normalisation)

$$\varpi_\alpha(\zeta) \propto \mu_\alpha(\zeta) q^{N(N-1)} (\alpha q^{2N-2|\Lambda|}; q^2)_\infty, \quad g_\alpha(\zeta) \propto q^{2N(N-1)} (\alpha q^{2N-2|\Lambda|}; q^2)_\infty (\alpha q^{-2N}; q^2)_\infty, \quad (9)$$

with window $\alpha \in (0, q^{2|\Lambda|})$, and let $\bar{D}$ denote (7) read with the *process* configuration in the N^- slot and the dual in the N^+ slot,

$$\bar{D}(\xi, \eta) = \prod_{i=1,2} \prod_{x: \xi_{i,x}=\eta_{i,x}=1} \left(1 - \frac{q^{2(x - N_{x-1}^-(\eta_i) + N_{x+1}^+(\xi_i))}}{\alpha_i}\right);$$

transposing the interlacing of Theorem 2.6(ii), $\bar{D}$ is again a self-duality function. Then

$$\sum_\eta \varpi_{\alpha_1}(\eta_1) \varpi_{\alpha_2}(\eta_2) \bar{D}(\xi, \eta) \bar{D}(\xi', \eta) = \delta_{\xi, \xi'} a(\xi), \quad a(\xi) = \prod_{i=1,2} \frac{g_{\alpha_i}(\xi_i)}{\varpi_{\alpha_i}(\xi_i)}, \quad (10)$$

which we have verified to machine precision for the $n = \infty$ rates on two and three sites (off-diagonals and the norm formula). Three observations. (i) The reweighting in (9) depends only on the conserved particle numbers, so $\varpi := \varpi_{\alpha_1} \otimes \varpi_{\alpha_2}$ is again reversible, and Lemma 2.8 applies with $(\varpi, \bar{D}, a)$. (ii) In the orientation $\bar{D}$ the process enters through the *left* counts $N^-(\eta_i)$, which are almost surely finite under ν on $\mathbb{Z}$: the degeneration that affects (7) in infinite volume (where $q^{2N^+(\eta)} = 0$ a.s.) does not occur — the resolution is to transpose the slots, not to reflect the lattice, which is impossible (the reflected generator fails the interlacing, consistently with the reflection

corresponding to the non-stochastic continuation $n < n_{\text{crit}}$ of Remark 2.3). (iii) In the regime (A) scaling, $q = 1 - c/N^2$ on $|\Lambda| = O(N)$ sites, the window is asymptotically $\alpha_i < 1$, i.e. both densities below $\frac{1}{2}$; the case of both densities above $\frac{1}{2}$ follows by the particle-hole-plus-reflection symmetry of (1) (which flips both species), while the mixed case $\rho_1 < \frac{1}{2} < \rho_2$ is *not* reached by the window and is deferred (e.g. via the signed-measure version of the orthogonality noted in [5, Rmk. after Thm. 3.2]). The reweighting is bounded above and below by constants (Lemma 6.5). The remaining infinite-volume step — replacing $[-KN, KN]$ by $\mathbb{Z}$ in the correlation functions — is deferred to the final version. In regime (B) we use the triangular duality of Corollary 2.7 instead. Lemma 2.8 reduces every space-time correlation of degree- k fields to a single k -particle dual kernel evaluation — the orthogonal-duality reduction of few-body correlations of [1, 2], and the mechanism that makes the fluctuation theory a few-body problem. In this paper it is applied at $k = 2$, to the inter-species cross-noise (§6).

3 Exact current decoupling and vanishing transport coefficients

The structural fact behind every decoupling result in this paper is elementary and exact, holding at the level of the microscopic rates: at $n = \infty$ each species' current does not see the other. We prove this first (Proposition 3.1), then draw its two consequences — that each marginal is an autonomous single-species ASEP, and that the two equilibrium cross-transport coefficients vanish identically (Proposition 3.3). Everything afterwards is the question of what residual coupling can still survive once these channels are closed.

Proposition 3.1 (exact current decoupling). *At $n = \infty$ the microscopic species- i current across a bond $(x, x+1)$ equals the single-species ASEP current in η_i alone,*

$$j_i = r_R \eta_{i,x} (1 - \eta_{i,x+1}) - r_L (1 - \eta_{i,x}) \eta_{i,x+1}, \quad r_R = 1, \quad r_L = q^2, \quad (11)$$

with no dependence on the other species. Consequently

- (a) *each marginal density field evolves as an autonomous single-species ASEP;*
- (b) *all cross derivatives of the macroscopic flux vanish identically, $\partial_{\rho_1} j_2 = \partial_{\rho_1}^2 j_2 = \partial_{\rho_1} \partial_{\rho_2} j_2 = 0$, so the two-species hydrodynamic flux Hessian is diagonal;*
- (c) *the inter-species coupling is carried entirely by correlated jumps — bound-pair translations, swaps, and binding/splitting.*

Proof. The species-1 current across $(x, x+1)$ is the signed rate of species-1 transfer, $j_1 = \sum_{\text{transitions}} r(\cdot) \Delta \eta_1$, summed over the bond transitions of (1). We collect the rightward part (the four configurations with $\eta_{1,x} = 1$, $\eta_{1,x+1} = 0$) and the leftward part (the four with $\eta_{1,x} = 0$, $\eta_{1,x+1} = 1$), in each case running over every species-2 occupancy of the bond.

Rightward. $(1, 0) \rightarrow (0, 1)$ at rate 1; $(3, 0) \rightarrow (0, 3)$ (pair hop) at rate 1; $(3, 2) \rightarrow (2, 3)$ (species 1 sliding past the pair) at rate 1; and from $(1, 2)$, species 1 leaves both by the merge $(1, 2) \rightarrow (0, 3)$ at rate ε and by the swap $(1, 2) \rightarrow (2, 1)$ at rate q^2 , summing to $\varepsilon + q^2 = 1$. All four give rightward rate $1 = r_R$, independently of the species-2 occupant.

Leftward. $(0, 1) \rightarrow (1, 0)$ at rate q^2 ; $(2, 3) \rightarrow (3, 2)$ at rate q^2 ; the swap $(2, 1) \rightarrow (1, 2)$ at rate q^2 ; and from $(0, 3)$, species 1 reaches x both by the pair hop $(0, 3) \rightarrow (3, 0)$ at rate q^4 and by the split $(0, 3) \rightarrow (1, 2)$ at rate $q^2 \varepsilon$, summing to $q^4 + q^2 \varepsilon = q^4 + q^2(1 - q^2) = q^2$. All four give leftward rate $q^2 = r_L$. (The complementary merge/split channels $(2, 1) \rightarrow (0, 3)$, $(0, 3) \rightarrow (2, 1)$ and the mixed

hop $(3, 1) \leftrightarrow (1, 3)$ leave η_1 on the bond unchanged and do not contribute.) Thus the species-2 occupant never changes the species-1 transfer rate, giving (11); the species-2 current is symmetric. Claim (a) is immediate. For (b), the hydrodynamic flux is $\mathbb{E}_{\nu_\rho}[j_i] = r_R \rho_i(1 - \rho_i) - r_L(1 - \rho_i)\rho_i = (1 - q^2)\rho_i(1 - \rho_i)$, a function of ρ_i alone, so every ρ_j -derivative with $j \neq i$ vanishes, as does $\partial_{\rho_1}\partial_{\rho_2}j_i$. Claim (c) is the residual: the only transitions correlating the two species are the bound-pair hops, swaps and split/merge events, none of which alters (11). $\square$

Proposition 3.2 (current decoupling at every n). *For every finite n , with the rates of [3, §2.1] (constructed in [12] for $n = 2, 3$), the species- i transfer rates across a bond are*

$$r_R = q^{-1}\beta_n, \quad r_L = q\beta_n, \quad \beta_n = q^{1-2n} + q^{2n-1},$$

independently of the species- j occupancy, $j \neq i$. Hence each marginal is an autonomous ASEP for every n , and (1) is the entrywise q^{2n} -rescaled $n \rightarrow \infty$ limit of the finite- n generator.

Proof. At finite n four channels are present that vanish at $n = \infty$: the left merges $(1, 2) \rightarrow (3, 0)$, $(2, 1) \rightarrow (3, 0)$, of rate $2 - q^{2n-2}(1 - q^2)$, and the left splits $(3, 0) \rightarrow (1, 2)$, $(3, 0) \rightarrow (2, 1)$, of rate $q^{-2}(2 - q^{2n-2}(1 - q^2))$. With $\sigma_n := (q^{n-1} - q^{1-n})^2 = q^{2n-2} - 2 + q^{2-2n}$, the species-1 rightward rate, case by case in the species-2 bond occupancy, reads

$$\begin{aligned} (1, 0): & \quad q^{-1}\beta_n = q^{-2n} + q^{2n-2}; \\ (3, 0): & \quad \underbrace{q^{-2}\sigma_n}_{(3,0) \rightarrow (0,3)} + \underbrace{q^{2n-2} - q^{2n-4} + 2q^{-2}}_{(3,0) \rightarrow (2,1)} = q^{-2n} + q^{2n-2}; \\ (1, 2): & \quad \underbrace{2 + q^{-2n}(1 - q^2)}_{(1,2) \rightarrow (0,3)} + \underbrace{\sigma_n}_{(1,2) \rightarrow (2,1)} = q^{-2n} + q^{2n-2}; \\ (3, 2): & \quad q^{-1}\beta_n, \end{aligned}$$

and the leftward rates all equal $q\beta_n = q^{2-2n} + q^{2n}$: in the $(0, 3)$ background, the pair hop $q^2\sigma_n$ plus the split $2q^2 + q^{2-2n} - q^{4-2n}$; in the $(2, 1)$ background, the swap σ_n plus the left merge $2 - q^{2n-2}(1 - q^2)$. Species 2 is symmetric. We have additionally verified the full statement, and the rescaled $n \rightarrow \infty$ limit, by computer algebra on the 16×16 generator of [3, §2.1]. $\square$

For the transport coefficients introduce the bond function

$$V_x := [\mathbf{1}_{(3,0)} + q^4\mathbf{1}_{(0,3)} - q^2\mathbf{1}_{(1,2)} - q^2\mathbf{1}_{(2,1)}]_{x,x+1}, \quad \mathbf{1}_{(a,b)} = \mathbf{1}\{\omega_x = a, \omega_{x+1} = b\}, \quad (12)$$

the carré du champ of the only events moving *both* species across the bond (bound-pair hops, sign +; swaps, sign -).

Proposition 3.3 (vanishing cross coefficients). $\mathbb{E}_\nu[V_x] = 0$ (equivalently, the equilibrium cross-mobility $\sigma_{12} = 0$), and the static cross-compressibility $C_{12} := \sum_r \langle \delta\rho_1(0)\delta\rho_2(r) \rangle_\nu = \text{Cov}(N_1, N_2)/L$ vanishes identically.

Proof. By Lemma 2.2, $\mathbb{E}_\nu[V_x] = \nu(3, 0) + q^4\nu(0, 3) - q^2\nu(1, 2) - q^2\nu(2, 1) = \nu(3, 0)[1 + 1 - 1 - 1] = 0$. The compressibility statement is the product structure of ν : the two species are independent under ν , so all equal-time cross-correlations vanish and $C_{12} = 0$. $\square$

Remark 3.4 (interpretation). By Proposition 3.1 the asymmetry — and hence the KPZ nonlinearity — lives only in the marginals, which are autonomous ASEPs, while the inter-species coupling is purely a noise effect; by Proposition 3.3 that coupling is invisible to both equilibrium transport

coefficients. Physically, a bound pair carries one particle of each species together, but it hops with exactly the bias the two single-species currents already have, so it injects correlated fluctuations without shifting any mean flow: coupling in the noise, not in the drift. These two facts underlie the decoupling in both regimes. The cancellation $1 + 1 - 1 - 1 = 0$ of Proposition 3.3 recurs three times in §6.

4 Local central limit theorems for the dual coordinates

The dual processes below reduce, after passing to a relative coordinate, to one-dimensional walks. We collect the local CLTs they require. Throughout, a *driftless finite-range walk* on $\mathbb{Z}$ means a continuous-time, irreducible, aperiodic, mean-zero walk with bounded jump range.

Lemma 4.1 (free and perturbed kernel bounds). *Let R^0 be the homogeneous symmetric nearest-neighbour walk of total rate $a > 0$. Then $\mathbb{P}(R^0(t) = r) \leq C/\sqrt{1+t}$ for all $t \geq 0$, $r \in \mathbb{Z}$, with $C = C(a)$. Moreover, let $\tilde{R}$ be an irreducible walk on $\mathbb{Z}$ (or on a state space graph-isomorphic to $\mathbb{Z}$, such as $\mathbb{Z} \setminus \{0\}$ relabelled) which is reversible with respect to a measure m with $c_1 \leq m \leq c_2$, has jump range $\leq \varrho$ and total exit rates $\leq \Lambda$, and whose nearest-neighbour conductances are bounded below, $m(x)r(x, x+1) \geq \delta > 0$ for all x . Then*

$$\sup_r \mathbb{P}(\tilde{R}(t) = r) \leq \frac{C}{\sqrt{1+t}}, \quad C = C(c_1, c_2, \delta, \Lambda, \varrho). \quad (13)$$

Every perturbed walk in this paper satisfies the hypotheses: the no-merge relative walk on $\mathbb{Z} \setminus \{0\}$ of Lemmas 4.2 and 7.3 is reversible with respect to counting measure (the swap channel is symmetric), with $\delta \geq q^2$, so its constants are uniform as $q = q_N \rightarrow 1$; the full relative walk of §8 is reversible with respect to $m(0) = q^{-2}$, $m \equiv 1$ otherwise, with $\delta = \min(\varepsilon, q^2) > 0$ at fixed q .

Proof. For R^0 , Fourier inversion gives $\mathbb{P}(R^0(t) = r) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-at(1-\cos\theta)} e^{-ir\theta} d\theta \leq \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-at(1-\cos\theta)} d\theta$. For $t \leq 1$ this is $\leq 1 \leq \sqrt{2}/\sqrt{1+t}$. For $t \geq 1$, using $1 - \cos\theta \geq \frac{2}{\pi^2}\theta^2$ on $[-\pi, \pi]$, $\int_{-\pi}^{\pi} e^{-at(1-\cos\theta)} d\theta \leq \int_{-\infty}^{\infty} e^{-\frac{2a}{\pi^2}t\theta^2} d\theta = \pi\sqrt{\frac{\pi}{2at}} \leq C/\sqrt{t}$.

For $\tilde{R}$, the bound is trivial for $t \leq 1$, so let $t \geq 1$. On $\ell^2(m)$ the generator is self-adjoint with Dirichlet form $\mathcal{E}(f) = \frac{1}{2} \sum_{x,y} m(x)r(x,y)(f(y) - f(x))^2 \geq \delta \sum_x (f(x+1) - f(x))^2$. On $\mathbb{Z}$ the discrete Nash inequality $\|f\|_2^6 \leq 4\|\nabla f\|_2^2 \|f\|_1^4$ holds: from $f(x)^2 = \sum_{y < x} \nabla(f^2)(y)$ and Cauchy-Schwarz, $\|f\|_\infty^2 \leq 2\|\nabla f\|_2 \|f\|_2$, so $\|f\|_2^2 \leq \|f\|_1 \|f\|_\infty \leq \|f\|_1 (2\|\nabla f\|_2 \|f\|_2)^{1/2}$, and rearranging gives the claim. With the measure bounds $c_1 \leq m \leq c_2$ this transfers to $\|f\|_{2,m}^6 \leq C\mathcal{E}(f) \|f\|_{1,m}^4$, a Nash inequality with dimension parameter $d = 1$ for the Dirichlet form of $\tilde{R}$. By the equivalence of Nash inequalities and on-diagonal kernel bounds for symmetric Markov semi-groups [6], $\sup_{x,y} p_t(x,y)/m(y) \leq Ct^{-1/2}$, and $m \leq c_2$ gives (13); all constants depend only on $(c_1, c_2, \delta, \Lambda, \varrho)$. $\square$

The Edwards-Wilkinson kernel requires, in addition, control of the slow sojourn at the origin created by the small split rate. Recall $\varepsilon = 1 - q^2$ and set $\nu_{\text{sp}} := 2q^2\varepsilon$. The relevant relative walk (defined in §5) agrees with a rate- $(1 + q^2)$ symmetric walk off $\{-1, 0, 1\}$, has a sticky origin held for an $\text{Exp}(\nu_{\text{sp}})$ time, and is re-entered at the origin only from ± 1 at rate ε .

Lemma 4.2 (defected marginal local CLT). *Let R be the walk just described, started at $R(0) = 0$. For $\nu_{\text{sp}}t \leq K$,*

$$\mathbb{P}(R(t) = r) \leq \frac{C}{\sqrt{1+t}} + \delta_{r,0} e^{-\nu_{\text{sp}}t}, \quad C = C(K, q).$$

Proof. Write $\nu = \nu_{\text{sp}}$, so $\varepsilon = \nu/(2q^2)$, and recall $\nu t \leq K$, $\nu \leq 2$; all constants are $C = C(K, q)$. For $t \leq 1$ the bound is trivial, so let $t \geq 1$. Three walks appear: R itself; the *no-merge* walk $\widehat{R}$ on $\mathbb{Z} \setminus \{0\}$ (the rates of R with the merge channel removed), which satisfies (13) with constants uniform in $q \rightarrow 1$; and the walk killed at the merges, whose kernel is dominated by that of $\widehat{R}$, since killing only removes mass: for $r \neq 0$ and any starting point in $\{\pm 1\}$,

$$\mathbb{P}_{\pm 1}(R(u) = r, \text{ no merge in } [0, u]) = \mathbb{E}_{\pm 1}[\mathbf{1}\{\widehat{R}(u) = r\} e^{-\varepsilon \Lambda_u}] \leq \frac{C}{\sqrt{1+u}}, \quad (14)$$

Λ_u the occupation of $\{\pm 1\}$ by $\widehat{R}$.

Atom. On $\{t < \tau_1\}$, $\tau_1 \sim \text{Exp}(\nu)$ the first split, the walk has never left 0: this is the term $\delta_{r,0} e^{-\nu t}$.

Occupation of the adjacent set. Let $g(t) := \int_0^t \mathbb{P}(R(u) \in \{\pm 1\}) du$. Exits from 0 occur at rate ν while at 0 (the two splits; pair hops do not change R), so the expected number of exits by time t is $\nu \int_0^t \mathbb{P}(R(u) = 0) du \leq \nu t \leq K$. By the strong Markov property at each exit and (14), each excursion contributes to g at most $\int_0^t C(1+v)^{-1/2} dv \leq C\sqrt{t}$ in expectation, whence $g(t) \leq (1+K)C\sqrt{t}$.

Off the origin. Fix $r \neq 0$. On $\{R(t) = r\}$ the time t lies in an excursion, begun at an exit from 0; the exit times form a point process of intensity $\nu \mathbf{1}\{R(u) = 0\} du$, and given an exit at u the probability that its excursion still runs at time t with $R(t) = r$ is bounded by (14). Hence

$$\mathbb{P}(R(t) = r) \leq \nu \int_0^t \mathbb{P}(R(u) = 0) \frac{C}{\sqrt{1+t-u}} du,$$

and we insert $\mathbb{P}(R(u) = 0) \leq e^{-\nu u} + \varepsilon g(u) \leq e^{-\nu u} + C\varepsilon\sqrt{u}$ (the two events being “never left” and “last merge before u ”, the latter of intensity $\varepsilon \mathbf{1}\{R \in \{\pm 1\}\}$). The exponential part: on $u \leq t/2$ it is at most $C(1+t)^{-1/2} \int_0^\infty \nu e^{-\nu u} du = C(1+t)^{-1/2}$; on $u > t/2$ it is at most $\nu e^{-\nu t/2} \cdot 2\sqrt{t} \leq C(1+t)^{-1/2}$, since $\nu\sqrt{t}\sqrt{1+t} \leq 2\nu(1+t) \leq 2(\nu+K)$. The occupation part is at most $C\nu\varepsilon \int_0^t \sqrt{u}(1+t-u)^{-1/2} du \leq C\nu\varepsilon\sqrt{t} \cdot 2\sqrt{t} = C\nu\varepsilon t$, and multiplying by $\sqrt{1+t}$, $\nu\varepsilon t\sqrt{1+t} \leq 2\nu\varepsilon t^{3/2} = 2(\nu t) \cdot \varepsilon\sqrt{t} = 2(\nu t) \cdot \frac{\sqrt{\nu}\sqrt{\nu t}}{2q^2} \leq \frac{\sqrt{2}K^{3/2}}{q^2} \leq C$, so this part too is $\leq C(1+t)^{-1/2}$.

At the origin. Decomposing $\{R(t) = 0\}$ on the last merge before t (or none),

$$\mathbb{P}(R(t) = 0) = e^{-\nu t} + \varepsilon \int_0^t \mathbb{P}(R(u) \in \{\pm 1\}) e^{-\nu(t-u)} du \leq e^{-\nu t} + \varepsilon \int_0^t \frac{C}{\sqrt{1+u}} du \leq e^{-\nu t} + C\varepsilon\sqrt{t},$$

using the off-origin bound just proved at $r = \pm 1$; and $\varepsilon\sqrt{t}\sqrt{1+t} \leq 2\varepsilon t = \nu t/q^2 \leq K/q^2$ shows $C\varepsilon\sqrt{t} \leq C(1+t)^{-1/2}$. Collecting the three displays gives the claim. $\square$

The relative coordinate is now controlled. The sum coordinate is an additive functional of the same event stream, so it needs a separate argument: conditionally on the unsigned event skeleton, the remaining randomness of S is a sum of independent non-degenerate signs, and we recover spreading from a concentration inequality rather than from a Fourier estimate.

Lemma 4.3 (conditional concentration for the sum coordinate). *Let S be the sum coordinate of §5, and let $\mathfrak{S}$ denote the unsigned skeleton: the path of R together with the times of the pair hops (the events with $\Delta R = 0$, which the R -path alone does not reveal), but not their signs. Let $M = M(\mathfrak{S})$ be the number of ambiguous events, those for which the skeleton admits two values of ΔS . Then for every $s' \in \mathbb{Z}$,*

$$\mathbb{P}(S(t) = s' \mid \mathfrak{S}) \leq \frac{C}{\sqrt{1+M}},$$

and M stochastically dominates a Poisson variable of mean t , so $\mathbb{P}(M < t/2) \leq e^{-ct}$.

Proof. Given $\mathfrak{S}$, $S(t) = m(\mathfrak{S}) + \sum_{j=1}^M \eta_j$ with $m(\mathfrak{S})$ determined by the skeleton and the η_j conditionally independent: on $\{|R| \geq 2\}$, and on the outward steps from $\{\pm 1\}$, a step $\Delta R = +1$ is either “species 1 left” ($\Delta S = -1$, conditional probability $q^2/(1+q^2)$) or “species 2 right” ($\Delta S = +1$), and symmetrically for $\Delta R = -1$; a pair hop has $\Delta S = \pm 2$ with conditional probabilities $(1, q^4)/(1+q^4)$. (Merges, splits and swaps determine ΔS and are unambiguous.) Each η_j is non-degenerate with both values of probability in $[\delta, 1-\delta]$, $\delta = \delta(q) > 0$, so by the Kolmogorov–Rogozin concentration inequality [15, Ch. III], $\mathbb{P}(S(t) = s' \mid \mathfrak{S}) \leq C(\sum_j (1 - Q(\eta_j; 0)))^{-1/2} \wedge 1 \leq C(1 + \delta M)^{-1/2}$. For the domination: in every state the total rate of ambiguous events is ≥ 1 — rate $2(1+q^2)$ on $\{|R| \geq 2\}$, $1+q^2$ (the outward steps) at $\{\pm 1\}$, and $1+q^4$ (the pair hops) at the origin — so, conditionally on any past, ambiguous events fire at rate ≥ 1 , M dominates a rate-1 Poisson count, and $\mathbb{P}(M < t/2) \leq e^{-ct}$. $\square$

The fixed- q analysis of §8 requires, beyond the upper bound (13), the exact leading occupation asymptotics of the relative walk, including the invariant-measure weighting at the defect. The following elementary lemma supplies them; its entire content is two exact identities (detailed balance and first-passage decomposition) plus the free first-passage transform.

Lemma 4.4 (occupation-time asymptotics). *Let X be an irreducible, recurrent walk on $\mathbb{Z}$, reversible with respect to a measure m with $c_1 \leq m \leq c_2$ and $m \equiv 1$ off a finite set, and agreeing off a finite set with the symmetric nearest-neighbour walk of rate a per direction. Then for every fixed r ,*

$$\tau_r(s) := \int_0^s \mathbb{P}_0(X(t) = r) dt \sim m(r) \sqrt{\frac{s}{\pi a}} \quad (s \rightarrow \infty).$$

Proof. Let $G_\lambda(x, y) = \int_0^\infty e^{-\lambda t} \mathbb{P}_x(X(t) = y) dt$, $\lambda > 0$. Two exact identities: detailed balance gives $m(x)G_\lambda(x, y) = m(y)G_\lambda(y, x)$, and the first-passage decomposition at y gives $G_\lambda(x, y) = \mathbb{E}_x[e^{-\lambda T_y}] G_\lambda(y, y)$. Combining,

$$G_\lambda(0, r) = \frac{m(r)}{m(0)} \mathbb{E}_r[e^{-\lambda T_0}] G_\lambda(0, 0), \quad (15)$$

and by recurrence $\mathbb{E}_r[e^{-\lambda T_0}] \rightarrow 1$ as $\lambda \downarrow 0$ for each fixed r . To anchor $G_\lambda(0, 0)$, sum (15) over r and use $\sum_r G_\lambda(0, r) = \lambda^{-1}$:

$$\frac{1}{\lambda} = \frac{G_\lambda(0, 0)}{m(0)} \sum_r m(r) \mathbb{E}_r[e^{-\lambda T_0}].$$

Let I be a finite interval containing the defect and the origin. From r outside I the walk is free until it first hits I (necessarily at the near edge), and from ∂I it reaches 0 in a time which is almost surely finite, so $\mathbb{E}_r[e^{-\lambda T_0}] = \varrho(\lambda)^{\text{dist}(r, I)} \mathbb{E}_{\partial I}[e^{-\lambda T_0}]$ with $\mathbb{E}_{\partial I}[e^{-\lambda T_0}] \rightarrow 1$, where $\varrho(\lambda)$ is the free nearest-neighbour first-passage root, $\lambda + 2a = a(\varrho + \varrho^{-1})$, i.e. $\varrho(\lambda) = 1 - \sqrt{\lambda/a} + O(\lambda)$. Hence $\sum_r m(r) \mathbb{E}_r[e^{-\lambda T_0}] = (2\sqrt{a/\lambda})(1 + o(1))$ and $G_\lambda(0, 0) = \frac{1}{2} m(0) (a\lambda)^{-1/2} (1 + o(1))$. Returning to (15), $G_\lambda(0, r) \sim \frac{1}{2} m(r) (a\lambda)^{-1/2}$. Since τ_r is the integral of a nonnegative function whose Laplace transform is regularly varying of index $-\frac{1}{2}$ at 0, Karamata’s Tauberian theorem gives $\tau_r(s) \sim \frac{1}{2} m(r) a^{-1/2} s^{1/2} / \Gamma(\frac{3}{2}) = m(r) \sqrt{s/(\pi a)}$. $\square$

5 The two-particle dual kernel bound

The order-two fields of the fluctuation analysis reduce, by Lemma 2.8, to the two-particle dual kernel. A bound pair carries one particle of *each* species, so two particles of the same species can never bind; the different-species pair is the bind/split/merge process. We treat the two channels.

Lemma 5.1 (same-species channel). *The same-species two-particle dual kernel obeys $p_t(\xi, \xi') \leq C/(1+t)$.*

Proof. With no binding move available, the two same-species dual particles perform a two-particle ASEP, for which the relative coordinate (a walk on $\mathbb{Z}_{\geq 1}$, free off the exclusion boundary) and the sum coordinate admit exactly the arguments of Lemmas 4.1 and 4.3: the relative-coordinate marginal is bounded by $C/\sqrt{1+t}$ (reversible, uniformly elliptic nearest-neighbour after the boundary defect), and conditionally on the unsigned skeleton the sum coordinate concentrates no faster than $C/\sqrt{1+t}$, giving the product bound. (Alternatively, the bound can be read off the exact Bethe-ansatz solution of the two-particle master equation [18].) The lemma is included for completeness: the cross-noise analysis of §6 uses only the different-species channel. $\square$

For the different-species pair with positions X_1, X_2 set

$$S := X_1 + X_2, \quad R := X_2 - X_1. \quad (16)$$

Off $\{-1, 0, 1\}$ the coordinate R is a symmetric nearest-neighbour walk of rate $1 + q^2$; the origin $R = 0$ (the bound pair) is sticky with split (escape) rate $\nu_{\text{sp}} = 2q^2\varepsilon$, entered from ± 1 only by a merge of rate ε . The coordinate S is an additive functional of the R -path. This is exactly the process of Lemmas 4.2–4.3.

Theorem 5.2 (type D two-particle kernel bound). *For the different-species dual started from a bound pair, in the window $\nu_{\text{sp}}t \leq K$,*

$$p_t(\xi, \xi') \leq \frac{C}{1+t} + e^{-\nu_{\text{sp}}t} \frac{C}{\sqrt{1+t}}. \quad (17)$$

Proof. Write $p_t(\xi, \xi') = \mathbb{E}[\mathbf{1}\{R(t) = r'\} \mathbb{P}(S(t) = s' \mid \mathfrak{S})]$ and split on $\{M \geq t/2\}$. There Lemma 4.3 gives $\mathbb{P}(S(t) = s' \mid \mathfrak{S}) \leq C/\sqrt{1+t}$, and taking the expectation against $\mathbf{1}\{R(t) = r'\}$ and applying Lemma 4.2 yields $(C/\sqrt{1+t} + \delta_{r',0} e^{-\nu_{\text{sp}}t}) \cdot C/\sqrt{1+t}$. On $\{M < t/2\}$ the contribution is at most $\mathbb{P}(M < t/2) \leq e^{-ct} \leq C/(1+t)$, absorbed into the first term of (17). $\square$

Remark 5.3. The bound state propagates in one dimension (whence the $t^{-1/2}$) but is exponentially damped at the split rate $\nu_{\text{sp}} \sim 4c/N^2$; its time integral is finite, $\int_0^\infty e^{-\nu_{\text{sp}}t} t^{-1/2} dt = \sqrt{\pi/\nu_{\text{sp}}} \sim N/\sqrt{c} = o(N^2)$. By Proposition 6.9 below it enters the species fields with zero weight, so the departure from the ASEP bound $C/(1+t)$ is invisible to the limiting fields and surfaces only in the crossover of §7.

6 The decoupled Edwards–Wilkinson limit

We now prove the first decoupling theorem by the martingale–problem scheme for fluctuation fields: identify the drift as a Laplacian (D), the noise as white with the right variance (N), and — the point at which the two species could have interacted — show the cross-noise between them vanishes (X). The single-species inputs are classical; the work is in (X) and in the drift remainder of (D), the only places where two dual particles of different species can bind.

Fix $\rho_1, \rho_2 \in (0, 1)$ (equivalently $\alpha_1, \alpha_2 > 0$); we do *not* require the symmetric line. For $\varphi \in \mathcal{S}(\mathbb{R})$ define the fluctuation fields

$$Y_i^N(\varphi, t) = \frac{1}{\sqrt{N}} \sum_{x \in \mathbb{Z}} \varphi\left(\frac{x}{N}\right) (\eta_{i,x}(tN^2) - \rho_i), \quad i = 1, 2. \quad (18)$$

Across the $O(N)$ field window the blocking density tilts by $O(1/N)$, so ν is locally homogeneous at the densities ρ_i and the limiting coefficients are constant.

Theorem 6.1 (decoupled Edwards–Wilkinson limit). *Let $\eta_0 \sim \nu$, so that the process is stationary. Under $q = q_N = 1 - c/N^2$, the pair (Y_1^N, Y_2^N) is tight in $D([0, T]; \mathcal{S}'(\mathbb{R}))^2$ and every limit point (Y_1, Y_2) decouples: each Y_i is a stationary solution of the linear stochastic heat equation*

$$\partial_t Y_i = D \partial_x^2 Y_i + \sqrt{2\chi_i D} \partial_x \xi_i,$$

with $D = 1$ the (common) species diffusivity — the symmetric-part diffusivity $\frac{1}{2}(1 + q_N^2) \rightarrow 1$, produced by the gradient computation of Proposition 6.11 — and $\chi_i = \rho_i(1 - \rho_i)$ the species- i compressibility, and the driving space-time white noises ξ_1, ξ_2 have vanishing cross-correlation (the cross bracket $\langle M_1, M_2 \rangle$ is zero in the limit); there is no coupling between the species in either the drift or the noise. The limit law is unique.

By Dynkin’s formula [13, App. 1.5] each field is a semimartingale

$$Y_i^N(\varphi, t) = Y_i^N(\varphi, 0) + \int_0^t \Gamma_i^N(\varphi, s) ds + M_i^N(\varphi, t), \quad (19)$$

with drift $\Gamma_i^N(\varphi, t) = N^{1/2} \sum_x \varphi'(\frac{x}{N}) W_{i,x}(\eta(tN^2))$, $W_{i,x}$ the instantaneous species- i current across $(x, x+1)$, and M_i^N a martingale whose predictable brackets are the carré du champ. Theorem 6.1 follows from the martingale–problem scheme [13, Ch. 11] once we establish, for each φ, T :

$$(D) \quad \Gamma_i^N(\varphi, \cdot) \rightarrow D Y_i(\Delta\varphi, \cdot) \text{ in } L^2;$$

$$(N) \quad \langle M_i^N \rangle(\varphi, t) \rightarrow 2\chi_i D t \|\partial_x \varphi\|_2^2;$$

$$(X) \quad \langle M_1^N, M_2^N \rangle(\varphi, t) \rightarrow 0;$$

together with tightness (Aldous–Mitoma) and uniqueness of the linear (Ornstein–Uhlenbeck) martingale problem, both classical [13, Ch. 11].

6.1 Single-species Gaussianity

Lemma 6.2. *For each i , Y_i^N converges to the Gaussian Ornstein–Uhlenbeck Edwards–Wilkinson field, with the diagonal bracket of (N).*

Proof. By Proposition 3.1(a) the species- i marginal is an autonomous WASEP of right/left rates $1, q^2$, i.e. asymmetry $\gamma = 1 - q^2 = 2c/N^2 + O(N^{-4})$. This sits below the KPZ threshold, which under diffusive scaling occurs at asymmetry $N^{-1/2}$ [9]; the Euler/Burgers term is $O(\gamma) \rightarrow 0$, and the process is in the Edwards–Wilkinson regime. The equilibrium density fluctuations of a one-species exclusion process in this regime converge to the Gaussian Ornstein–Uhlenbeck solution of $\partial_t Y = D \partial_x^2 Y + \sqrt{2\chi_i D} \partial_x \xi$, $\chi_i = \rho_i(1 - \rho_i)$ — the classical one-species equilibrium fluctuation theorem [13, Ch. 11], established for orthogonal duality systems in [2] — with the diagonal bracket $2\chi_i D$. (The classical statements concern the homogeneous Bernoulli equilibrium; under the blocking measure the density tilts by $O(1/N)$ across the field window, and the standard comparison — coupling the dynamics from the tilted and untilted equilibria and bounding the discrepancy field in L^2 — transfers the result. We omit the routine details.) $\square$

The content is thus in (X) and (D), where two dual particles of different species can bind.

6.2 The cross noise

The predictable cross bracket is $\langle M_1^N, M_2^N \rangle(\varphi, t) = \int_0^t \Theta^N \, ds$,

$$\Theta^N = \frac{1}{N} \sum_x \varphi' \left(\frac{x}{N} \right)^2 V_x, \quad (20)$$

V_x as in (12). By Proposition 3.3, $\mathbb{E}_\nu[\Theta^N] = 0$.

Lemma 6.3 (orthogonality to the density fields; density-free). *For all $\alpha_1, \alpha_2 > 0$ (in particular off the symmetric line), every $i \in \{1, 2\}$ and every $y \in \mathbb{Z}$, $\langle V_x, \eta_{i,y} - \rho_{i,y} \rangle_\nu = 0$, where $\rho_{i,y}$ is the local density of species i at y . Hence V_x has no order-one component in the local-basis decomposition (6); its lowest order is two.*

Proof. Since $\mathbb{E}_\nu[V_x] = 0$ (Proposition 3.3, an identity in α_1, α_2),

$$\langle V_x, \eta_{i,y} - \rho_{i,y} \rangle_\nu = \mathbb{E}_\nu[V_x \eta_{i,y}] - \rho_{i,y} \mathbb{E}_\nu[V_x] = \mathbb{E}_\nu[V_x \eta_{i,y}],$$

so it suffices to show $\mathbb{E}_\nu[V_x \eta_{i,y}] = 0$. For $y \notin \{x, x+1\}$ this is immediate: V_x is supported on the bond $\{x, x+1\}$ and ν is a product, so $\eta_{i,y}$ is independent of V_x and $\mathbb{E}_\nu[V_x \eta_{i,y}] = \mathbb{E}_\nu[V_x] \rho_{i,y} = 0$. For $y \in \{x, x+1\}$, the detailed-balance relations (4) — $\nu(0, 3) = q^{-4} \nu(3, 0)$, $\nu(1, 2) = \nu(2, 1) = q^{-2} \nu(3, 0)$, identities in the fugacities — give the ν -weighted values of V_x on its four support configurations as $\nu(3, 0) \cdot 1$, $\nu(0, 3) \cdot q^4$, $\nu(1, 2) \cdot (-q^2)$, $\nu(2, 1) \cdot (-q^2) = \nu(3, 0)(+1, +1, -1, -1)$. Hence for any bond function f , $\mathbb{E}_\nu[V_x f] = \nu(3, 0)[f(3, 0) + f(0, 3) - f(1, 2) - f(2, 1)]$, and each single-site occupation equals 1 on exactly one “+” and one “−” configuration:

	(3, 0)	(0, 3)	(1, 2)	(2, 1)
$\eta_{1,x}$	1	0	1	0
$\eta_{1,x+1}$	0	1	0	1
$\eta_{2,x}$	1	0	0	1
$\eta_{2,x+1}$	0	1	1	0

so $f(3, 0) + f(0, 3) - f(1, 2) - f(2, 1) = 0$ for each row, giving $\mathbb{E}_\nu[V_x \eta_{i,y}] = 0$. The mechanism is the $1 + 1 - 1 - 1$ cancellation of Proposition 3.3 one order higher; it uses only detailed balance, not species exchange, and is therefore density-free.

Orthogonality and support in the exact pairing. The expansion of V_x used below is in the family $\bar{D}$ with the pairing $\langle \cdot, \cdot \rangle_\varpi$ of Remark 2.9; since the mass-one functions $\bar{D}$ are nonlocal, the polynomial statement above does not by itself locate the expansion, and we check directly. The key structural fact: on the four support configurations of V_x the bond carries exactly one particle of each species, so the particle numbers (N_1, N_2) — hence the sector-reweighting factors of (9) — agree on all four, and the master identity $\mathbb{E}_\varpi[V_x f] = \sum_r \tilde{\varpi}(r) \nu(3, 0)[f(3, 0) + f(0, 3) - f(1, 2) - f(2, 1)](r)$ persists, with $\tilde{\varpi}$ the reweighted measure on the configuration r off the bond and the bracket evaluated at fixed r . The factor of $\bar{D}(\xi, \cdot)$ attached to a dual particle at w depends on η through $\mathbf{1}\{\eta_{i,w} = 1\}$ and $q^{-2N_{w-1}^-(\eta_i)}$: for $w < x$ it is independent of the bond; for $w > x+1$ its bond dependence is $q^{-2(\eta_{i,x} + \eta_{i,x+1})} \equiv q^{-2}$ on the support of V_x ; and for $w \in \{x, x+1\}$ its bond part is $\mathbf{1}\{\eta_{i,x}\}$ resp. $\mathbf{1}\{\eta_{i,x+1}\} q^{-2\eta_{i,x}}$, with values $(1, 0, 1, 0)$, $(1, 0, 0, 1)$ resp. $(0, 1, 0, 1)$, $(0, 1, 1, 0)$ on the four configurations, whose alternating sums vanish. Hence $\langle V_x, \bar{D}(\xi, \cdot) \rangle_\varpi = 0$ whenever ξ has at most one dual particle on the bond — in particular for all $|\xi| \leq 1$ — and also when the two bond particles are of the same species (the product of their indicators vanishes on the support of V_x). The expansion of V_x is therefore supported on ξ consisting of one different-species dual pair on the

bond, possibly dressed by further particles off the bond; numerically (four sites, machine precision) the dressing is moreover one-sided — every coefficient with a dual particle strictly to the *right* of the bond vanishes, reflecting the nested structure of the family (10), in which the factor of the rightmost dual particle is conditionally centred given the configuration to its left. $\square$

Lemma 6.4 (equal-time variance). $\mathbb{E}_\nu[(\Theta^N)^2] \leq C(\varphi) N^{-1}$.

Proof. $\mathbb{E}_\nu[(\Theta^N)^2] = N^{-2} \sum_{x,y} \varphi'(\frac{x}{N})^2 \varphi'(\frac{y}{N})^2 \mathbb{E}_\nu[V_x V_y]$. Each V_x has $\mathbb{E}_\nu[V_x] = 0$ and is supported on $\{x, x+1\}$; by the product structure $\mathbb{E}_\nu[V_x V_y] = 0$ unless $|x-y| \leq 1$. The $O(1)$ surviving near-diagonal terms per x are bounded, so the double sum is $\leq C \sum_x \varphi'(\frac{x}{N})^4 \leq CN$ (a Riemann sum). The prefactor N^{-2} gives the claim. $\square$

The expansion of Remark 2.9 controls correlations under the reweighted measure ϖ ; the following comparison transfers them to ν .

Lemma 6.5 (sector comparison). *On $\Lambda = [-KN, KN]$ with $q = 1 - c/N^2$ and α_1, α_2 fixed in $(0, 1)$, the measures ν and ϖ of Remark 2.9 have the same conditional laws given the conserved pair (N_1, N_2) , and their sector weights are comparable: $M := \sup_{N_1, N_2} \nu(N_1, N_2) / \varpi(N_1, N_2) \leq C(c, K, \alpha)$, uniformly in N . Consequently, for all $f, h \in L^2$ and $t \geq 0$,*

$$|\mathbb{E}_\nu[f(\eta_0) h(\eta_t)]| \leq M \mathbb{E}_\varpi[f(\eta_0) f(\eta_t)]^{1/2} \mathbb{E}_\varpi[h(\eta_0) h(\eta_t)]^{1/2}.$$

Proof. The two measures differ by the factor $h_1(N_1)h_2(N_2)$ of (9) (times constants), so the conditionals agree, and the ratio of sector weights is controlled by h_i : $q^{N(N-1)} \geq q^{|\Lambda|^2} = e^{-4cK^2(1+o(1))}$, and $1 \geq (\alpha q^{2N-2|\Lambda|}; q^2)_\infty \geq (\alpha'; q^2)_\infty > 0$ with $\alpha' = \alpha q^{-2|\Lambda|} = \alpha e^{4cK/N(1+o(1))} \leq \frac{1+\alpha}{2} < 1$ for N large; so the reweighting lies in $[C^{-1}, 1]$ and $M \leq C(c, K, \alpha)$. For the correlation bound, condition on the sector: the restricted semigroup is self-adjoint and positive with respect to the canonical measure (it is the square of the self-adjoint $P_{t/2}$), so $|\langle f, P_t h \rangle_{\text{can}}| \leq \langle f, P_t f \rangle_{\text{can}}^{1/2} \langle h, P_t h \rangle_{\text{can}}^{1/2}$. Summing over sectors with weights $\nu(N_1, N_2)$, Cauchy–Schwarz on the sector sum, and replacing the ν -weights by ϖ -weights at the cost of M (the summands $\langle f, P_t f \rangle_{\text{can}}$ being nonnegative) gives the claim. $\square$

Lemma 6.6 (the dressed mass is asymptotically negligible). *Write $V_z = V_z^{(2)} + V_z^{(\text{dr})}$ for the decomposition of V_z in the family (10) into its four different-species bond-pair components and the dressed remainder (Lemma 6.3). In the regime (A) scaling there is $\varepsilon_N \rightarrow 0$ with $\|V_z^{(\text{dr})}\|_{L^2(\varpi)}^2 \leq \varepsilon_N$, uniformly over z in the field window.*

Proof sketch; details deferred. Two ingredients. (i) The dressing is one-sided (Lemma 6.3): only particles strictly to the left of the bond occur. (ii) Over the field window the counting factors of $\bar{D}$ satisfy $q^{-2N_{w-1}^-(\eta_i)} = 1 + O(N^{-1})$ uniformly, so the family deviates from its homogeneous local limit — in which the dressed coefficients vanish identically — perturbatively. Numerically the dressed mass decays rapidly as $q \rightarrow 1$ at fixed window-scaled fugacities (the ratio of dressed to bond-pair mass falls from $7 \cdot 10^{-2}$ at $q = 0.7$ to $5 \cdot 10^{-5}$ at $q = 0.999$ on four sites); we expect $\varepsilon_N = O(N^{-1})$ or better, and only $\varepsilon_N \rightarrow 0$ is used below. $\square$

Proposition 6.7 (L^2 concentration of the cross bracket). $\mathbb{E}_\nu[\langle M_1^N, M_2^N \rangle(\varphi, t)^2] \leq C(\varphi, c) t (N^{-1} + N^{-2} \log(tN^2) + t\varepsilon_N) \rightarrow 0$, with ε_N as in Lemma 6.6; hence $\langle M_1^N, M_2^N \rangle(\varphi, t) \rightarrow 0$ in L^2 , establishing (X).

Proof. Let $C_\Theta(s) = \mathbb{E}_\nu[\Theta^N(\eta_0)\Theta^N(\eta_{sN^2})]$. By stationarity,

$$\mathbb{E}_\nu\left[\left(\int_0^t \Theta^N ds\right)^2\right] = 2 \int_0^t (t-s)C_\Theta(s) ds \leq 2t \int_0^t C_\Theta(s) ds. \quad (21)$$

Work on $\Lambda = [-KN, KN]$ in the setting of Remark 2.9, K fixed (the replacement of the infinite system by Λ is the deferred step recorded there). By Lemma 6.5,

$$|\mathbb{E}_\nu[V_x(\eta_0)V_y(\eta_u)]| \leq M G(x, u)^{1/2} G(y, u)^{1/2}, \quad G(z, u) := \mathbb{E}_\varpi[V_z(\eta_0)V_z(\eta_u)] \geq 0.$$

Split $V_z = V_z^{(2)} + V_z^{(\text{dr})}$ as in Lemma 6.6. The dual dynamics conserves the particle numbers $(|\xi_1|, |\xi_2|)$, so the mass sectors of the expansion are invariant under P_u and the cross terms vanish: $G = \mathbb{E}_\varpi[V^{(2)}P_u V^{(2)}] + \mathbb{E}_\varpi[V^{(\text{dr})}P_u V^{(\text{dr})}]$. For the first part, Lemma 2.8 (with $\varpi, \bar{D}, a$) gives $\mathbb{E}_\varpi[V_z^{(2)}P_u V_z^{(2)}] = \sum_{\xi, \xi'} c_\xi c_{\xi'} p_u(\xi, \xi') a(\xi')$, a sum over at most sixteen pairs of different-species bond configurations with coefficients and norms bounded in the window, each kernel entry bounded by Theorem 5.2; hence $\mathbb{E}_\varpi[V_z^{(2)}P_u V_z^{(2)}] \leq C[(1+u)^{-1} + e^{-\nu_{\text{sp}}u}(1+u)^{-1/2}]$. For the dressed part, P_u is an $L^2(\varpi)$ contraction, so $\mathbb{E}_\varpi[V_z^{(\text{dr})}P_u V_z^{(\text{dr})}] \leq \varepsilon_N$ by Lemma 6.6. With $u = sN^2$, $\nu_{\text{sp}}sN^2 = 4cs(1+o(1))$, and $\sum_x \varphi'(\frac{x}{N})^2 \leq C(\varphi)N$,

$$|C_\Theta(s)| \leq \frac{M}{N^2} \left(\sum_x \varphi'(\frac{x}{N})^2 G(x, sN^2)^{1/2} \right)^2 \leq C \left[\frac{1}{sN^2} + \frac{e^{-4cs}}{N\sqrt{s}} + \varepsilon_N \right] \quad (22)$$

for $s \geq N^{-2}$, while $|C_\Theta(s)| \leq CN^{-1}$ always (Lemma 6.4). Integrating, $\int_0^t C_\Theta \leq C(N^{-2}(1 + \log(tN^2)) + c^{-1/2}N^{-1} + t\varepsilon_N)$, and (21) gives the claim. $\square$

Remark 6.8. No Kipnis–Varadhan H_{-1} estimate enters: its role is played by Lemma 6.3. Without the mass ≤ 1 orthogonality the cross bracket would carry a slow density-type contribution with no useful time decay; with it, the correlation in (22) is carried by the two-particle kernel and integrates with at most a logarithmic loss. This is the recursive-martingale viewpoint of [2].

6.3 The drift

By Proposition 3.1 the species-1 current $W_{1,x}$ is a function of η_1 alone — the single-species ASEP current — which expands exactly, with $\gamma = 1 - q^2$ and $\nabla f_x = f_{x+1} - f_x$, as

$$W_{1,x} = -\nabla \eta_{1,x} + \gamma \eta_{1,x+1}(1 - \eta_{1,x}). \quad (23)$$

The first term is a discrete gradient; the second carries the prefactor $\gamma = O(N^{-2})$. The merge channel, in which species 1 binds onto a species-2 site, does *not* introduce an η_2 -dependence: by Proposition 3.1 the species-2 occupant never alters the species-1 transfer rate (the merge rate ε and the swap rate q^2 combine to the same 1 as the hop onto an empty site). Thus the drift involves only the autonomous single-species WASEP η_1 , and no two-species (bound-pair) structure enters it; the gradient part will give the Laplacian and the $O(\gamma)$ part is the switched-off KPZ nonlinearity. The bound-pair mode

$$B_z := (\eta_{1,z} - \rho_1)(\eta_{2,z} - \rho_2) \quad (24)$$

is the one near-conserved order-two object (it relaxes only at the split rate $\sim \varepsilon$); it does *not* enter the drift (23). We nonetheless record the following orthogonality, which makes the absence of any slow contribution explicit.

Corollary 6.9 (current orthogonal to the bound–pair mode). *For all $\rho_1, \rho_2 \in (0, 1)$, $\langle W_{i,x}, B_z \rangle_\nu = 0$ for all z , where $B_z = (\eta_{1,z} - \rho_1)(\eta_{2,z} - \rho_2)$.*

Proof. By Proposition 3.1 (equivalently (23)), $W_{1,x}$ is a function of η_1 alone. Since ν is a product measure with the two species independent and $\mathbb{E}_\nu[\eta_{2,z} - \rho_2] = 0$,

$$\langle W_{1,x}, B_z \rangle_\nu = \mathbb{E}_\nu[W_{1,x}(\eta_{1,z} - \rho_1)] \mathbb{E}_\nu[\eta_{2,z} - \rho_2] = 0,$$

and symmetrically for $W_{2,x}$. $\square$

Remark 6.10 (no off–symmetric–line obstruction). Because the overlap vanishes at *every* (ρ_1, ρ_2) (Corollary 6.9), the bound–pair mode never couples to the species current: there is no slow third hydrodynamic field, and hence no obstruction to the decoupling, off the symmetric line either. (A naive expansion of $W_{1,x}$ that misreads the merge channel as an η_2 –dependence produces a spurious overlap $\propto (\rho_1 - \rho_2)$; (23) shows it vanishes identically.)

Proposition 6.11 (drift). *For each i and any density $\rho_i \in (0, 1)$, $\Gamma_i^N(\varphi, \cdot) \rightarrow D Y_i(\Delta\varphi, \cdot)$ in L^2 , with D the diffusivity of the symmetric part of the single–species dynamics, by an elementary L^2 estimate. This establishes (D).*

Proof. By (23), $\Gamma_1^N = \Gamma_1^{N,\text{grad}} + \gamma \Gamma_1^{N,\text{cor}}$ with

$$\Gamma_1^{N,\text{grad}} = N^{1/2} \sum_x \varphi'(\frac{x}{N})(-\nabla \eta_{1,x}), \quad \Gamma_1^{N,\text{cor}} = N^{1/2} \sum_x \varphi'(\frac{x}{N}) g_x, \quad g_x := \eta_{1,x+1}(1 - \eta_{1,x}).$$

Gradient term. A summation by parts gives $\Gamma_1^{N,\text{grad}} = N^{1/2} \sum_x (\nabla \varphi')(\frac{x}{N}) \eta_{1,x} = N^{-1/2} \sum_x \varphi''(\frac{x}{N})(\eta_{1,x} - \rho_1) + o(1) = D Y_1^N(\Delta\varphi) + o(1)$ in L^2 , D the symmetric–part diffusivity; this is the standard gradient computation [13, Ch. 11]. Since the symmetric part of each species is simple symmetric exclusion — a gradient system — D is density–independent, hence the same constant for both species and at every (ρ_1, ρ_2) . *Correction term.* Write $g_x = \bar{g} + (g_x - \bar{g})$, $\bar{g} = \rho_1(1 - \rho_1)$. The mean part $\gamma N^{1/2} \bar{g} \sum_x \varphi'(\frac{x}{N})$ is the Euler flux $\partial_x[\gamma \rho_1(1 - \rho_1)]$, of order $O(\gamma) \rightarrow 0$ in the EW window. For the fluctuating part, write $F := \gamma N^{1/2} \sum_x \varphi'(\frac{x}{N})(g_x - \bar{g})$; by Cauchy–Schwarz in time and stationarity,

$$\mathbb{E}\left[\left(\int_0^t F(\eta(sN^2)) ds\right)^2\right] \leq t \int_0^t \mathbb{E}[F(\eta(sN^2))^2] ds = t^2 \mathbb{E}_\nu[F^2],$$

an *equal–time* second moment. Since ν is a product measure and $g_x - \bar{g}$ is a centred function of $\{\eta_{1,x}, \eta_{1,x+1}\}$, the covariance $\mathbb{E}_\nu[(g_x - \bar{g})(g_y - \bar{g})]$ vanishes unless $|x - y| \leq 1$, so $\mathbb{E}_\nu[F^2] = \gamma^2 N \sum_{x,y} \varphi' \varphi' \mathbb{E}_\nu[(g_x - \bar{g})(g_y - \bar{g})] \leq C \gamma^2 N \sum_x \varphi'(\frac{x}{N})^2 \leq C \gamma^2 N^2$, and the bound is $\leq t^2 \cdot C \gamma^2 N^2 = C c^2 N^{-2} t^2 \rightarrow 0$. Hence $\Gamma_1^N \rightarrow D Y_1(\Delta\varphi)$ in L^2 ; the species–2 drift is identical. $\square$

Granting (X), (N), (D), the limiting martingales (M_1, M_2) are continuous (jumps $\leq N^{-1/2} \|\varphi\|_\infty \rightarrow 0$) and Gaussian with diagonal bracket (Lemma 6.2) and vanishing cross bracket (Proposition 6.7); so the two driving noises are uncorrelated and the limit is the decoupled pair of EW fields of Theorem 6.1, unique by [13, Ch. 11]. $\square$

7 The initial–condition crossover

By Theorem 6.1 the two limiting fields are decoupled, so the coupling the model exhibits is not in the limiting dynamics. We locate it in the initial data and compute it. Run the different–species dual of §5 from a bound pair at the origin, with $q = 1 - c/T$, observed at time T . Write S, R as in (16).

Lemma 7.1 (split time). *The first split time τ is $\text{Exp}(\nu_{\text{sp}})$, $\nu_{\text{sp}} = 2q^2\varepsilon$, and $\nu_{\text{sp}}T \rightarrow 4c$, $\tau/T \Rightarrow V \sim \text{Exp}(4c)$, $\mathbb{P}(\tau > T) \rightarrow e^{-4c}$.*

Proof. While bound the pair undergoes only the rate 1 and q^4 hops (which preserve the bound phase) and the two splits (each rate $q^2\varepsilon$, leaving it); the bound phase is a single state whose only phase-changing exit is the split, so its holding time is $\text{Exp}(2q^2\varepsilon)$. The limits follow from $\varepsilon = 2c/T + O(T^{-2})$. $\square$

Lemma 7.2 (no re-binding). *The expected number of merges in $[\tau, T]$ is $O(c/\sqrt{T}) \rightarrow 0$.*

Proof. After the split, R is (away from a merge) a symmetric rate- $(1+q^2)$ walk; a merge occurs only from $R = \pm 1$, at rate ε . The expected occupation of $\{\pm 1\}$ up to T is $O(\sqrt{T})$ by Lemma 4.1 (with re-binding episodes handled by the excursion bound in the proof of Lemma 4.2, which contributes only the factor $1 + \nu_{\text{sp}}T = O(1+c)$), so the expected number of merges is $\leq \varepsilon \cdot O(\sqrt{T}) = O(c/\sqrt{T})$ for fixed c . $\square$

It is convenient to pass to the sum and difference $S := X_1 + X_2$, $R := X_2 - X_1$, so that $X_i = (S \mp R)/2$ and the interaction (confined to the adjacent set $\{|R| = 1\}$ and the origin) acts only on R .

Lemma 7.3 (exact symmetry identities and adjacent occupation). *Species-interchange symmetry maps $R \mapsto -R$, $S \mapsto S$ and fixes the law (the initial bound pair is symmetric); consequently*

$$\mathbb{E}[R(t)] = 0, \quad \text{Cov}(S(t), R(t)) = 0 \quad (\forall t \geq 0). \quad (25)$$

Moreover, on the event $A_T := \{\tau \leq T, \text{ no re-binding}\}$, the occupation $\Lambda_T := \int_\tau^T \mathbf{1}\{|R(t)| = 1\} dt$ of the adjacent set obeys $\mathbb{E}[\Lambda_T] = O(\sqrt{T})$.

Proof. The identities (25) follow from $X_i = (S \mp R)/2$ and the invariance of the law of the process under $R \mapsto -R$. After the split and with no merge, R is a continuous-time walk on $\mathbb{Z} \setminus \{0\}$ of total rate $\leq 2(1+q^2)$, symmetric on $\{|R| \geq 2\}$; by Lemma 4.1 its time- t marginal is $\leq C/\sqrt{1+t}$ uniformly, so the expected occupation of $\{\pm 1\}$ up to T is $\int_0^T C/\sqrt{1+t} dt = O(\sqrt{T})$. $\square$

Proposition 7.4 (two-phase functional CLT). *Conditionally on $\tau/T \rightarrow u \in [0, 1]$ and on A_T ,*

$$\frac{1}{\sqrt{2T}}(X_1(T), X_2(T)) \Rightarrow (\sqrt{u}\xi_0 + \sqrt{1-u}\xi_1, \sqrt{u}\xi_0 + \sqrt{1-u}\xi_2),$$

with ξ_0, ξ_1, ξ_2 i.i.d. $N(0, 1)$ — a standard bivariate normal of correlation u . Mixing over $\tau/T \Rightarrow U = \min(\text{Exp}(4c), 1)$ gives the limiting law of $(X_1, X_2)/\sqrt{2T}$.

Proof. (Conditioning on A_T is innocuous: $\mathbb{P}(A_T^c \mid \tau \leq T) \rightarrow 0$ by Lemma 7.2, so the conditioned and unconditioned post-split laws agree to vanishing total variation; we may therefore compute with the no-merge rates.) *Pre-split.* On $[0, \tau]$ the common pair position D is a walk of right rate 1 and left rate q^4 , hence variance rate $1+q^4 \rightarrow 2$ and drift $1-q^4 = O(c/T)$. By Donsker's invariance principle with $\tau \approx uT$, $D(\tau)/\sqrt{2T} \Rightarrow \sqrt{u}\xi_0$ (the drift contributes $(1-q^4)\tau/\sqrt{2T} = O(c\sqrt{u}/\sqrt{T}) \rightarrow 0$). Since $S(\tau) = 2D(\tau) + O(1)$ and $R(\tau) = O(1)$, the split deposits $S(\tau)/\sqrt{2T} \Rightarrow 2\sqrt{u}\xi_0$ and $R(\tau)/\sqrt{2T} \rightarrow 0$.

Post-split. Write $\bar{S}(s) = S(\tau+s) - S(\tau)$, $\bar{R}(s) = R(\tau+s) - R(\tau)$; each is a compensated jump process. Reading the rates of §5, the instantaneous drifts vanish for R on $\{|R| \geq 2\}$ and equal 2ε for S , while on $\{|R| = 1\}$ they are $O(\varepsilon)$ for R and ε for S (the deficit $2\varepsilon \rightarrow \varepsilon$ is the excluded merge); hence $\mathbb{E}[\bar{R}(T-\tau)]$ is supported on $\{|R| = 1\}$ with magnitude $\leq \varepsilon \mathbb{E}[\Lambda_T] = O(c/\sqrt{T})$, and

$\mathbb{E}[\bar{S}(T-\tau)] = 2\varepsilon(T-\tau) - \varepsilon \mathbb{E}[\Lambda_T] = O(c)$, both $o(\sqrt{T})$. The predictable quadratic-variation rates are, on $\{|R| \geq 2\}$,

$$a_S = a_R = 2(1 + q^2), \quad a_{SR} = 0,$$

and on $\{|R| = 1\}$ the bounded values $a_S = 1 + q^2$, $a_R = 1 + 5q^2$, $a_{SR} = \varepsilon$ (the swap $R = \pm 1 \mapsto \mp 1$ contributes $\Delta R = \mp 2$). Therefore

$$\langle \bar{S} \rangle = 2(1 + q^2)(T-\tau) - (1 + q^2)\Lambda_T, \quad \langle \bar{R} \rangle = 2(1 + q^2)(T-\tau) + (3q^2 - 1)\Lambda_T, \quad \langle \bar{S}, \bar{R} \rangle = \varepsilon \Lambda_T,$$

and by Lemma 7.3 every Λ_T -term is $O_{\mathbb{P}}(\sqrt{T}) = o_{\mathbb{P}}(T)$. With $q^2 \rightarrow 1$ and $(T-\tau)/T \rightarrow 1 - u$,

$$\frac{1}{2T} \langle \bar{S} \rangle \rightarrow 2(1 - u), \quad \frac{1}{2T} \langle \bar{R} \rangle \rightarrow 2(1 - u), \quad \frac{1}{2T} \langle \bar{S}, \bar{R} \rangle \rightarrow 0.$$

The jumps are bounded by 2, so the martingale functional CLT [8, Thm. 7.1.4] gives $(\bar{S}, \bar{R})(T-\tau)/\sqrt{2T} \Rightarrow (\sqrt{2(1-u)}\eta_S, \sqrt{2(1-u)}\eta_R)$ with η_S, η_R i.i.d. $N(0, 1)$, independent of $\mathcal{F}_\tau$ (hence of ξ_0) by the strong Markov property at τ .

Recombination. Adding the two phases, $S(T)/\sqrt{2T} \Rightarrow 2\sqrt{u}\xi_0 + \sqrt{2(1-u)}\eta_S$ and $R(T)/\sqrt{2T} \Rightarrow \sqrt{2(1-u)}\eta_R$. Setting $\xi_1 = (\eta_S - \eta_R)/\sqrt{2}$, $\xi_2 = (\eta_S + \eta_R)/\sqrt{2}$ (an orthogonal change of variables, so ξ_0, ξ_1, ξ_2 are i.i.d. $N(0, 1)$) and using $X_i = (S \mp R)/2$ gives the claim. Integrating against $\tau/T \Rightarrow U$ of Lemma 7.1 yields the U -mixture. $\square$

Theorem 7.5 (closed forms for the crossover). *With U as above and $g(u) = \mathbb{E}[Z_1^+ Z_2^+]$ the positive-part covariance of a standard bivariate normal of correlation u ,*

$$\text{Corr}(X_1, X_2) \rightarrow \frac{1 - e^{-4c}}{4c}, \quad \text{Corr}(X_1^+, X_2^+) \rightarrow \frac{\pi}{8(\pi - 1)c} [1 - 2e^{-4c} + I_0(4c) - L_0(4c)],$$

with I_0 the modified Bessel and L_0 the modified Struve function. Both tend to 1 as $c \rightarrow 0$ and decay as $1/c$ as $c \rightarrow \infty$.

Proof. Full position. By the exact identity $\text{Cov}(S, R) = 0$, $\text{Corr}(X_1, X_2) = (\text{Var } S - \text{Var } R)/(\text{Var } S + \text{Var } R)$. The per-state increment second moments of S, R satisfy $a_r + c_r = 4(1 + q^2)$ for every internal state r (a direct check of the rates), so $\text{Var } S + \text{Var } R = 4(1 + q^2) \sum_r \tau_r = 4(1 + q^2)T$ up to the drift-compensator variances, which are $O(1)$ (the compensators are $O(\varepsilon)$ -Lipschitz functionals of occupations) and are absorbed in the error terms below, with τ_r the occupation of state r ; and $a_r - c_r$ is supported on $\{0, \pm 1\}$ with $a_0 - c_0 = 4(1 + q^4)$, $a_{\pm 1} - c_{\pm 1} = -4q^2$, so $\text{Var } S - \text{Var } R = 4(1 + q^4)\tau_0 - 4q^2(\tau_{+1} + \tau_{-1})$. In the scaling limit the bound-state occupation τ_0 is the first-split time (re-binding negligible, Lemma 7.2), $\mathbb{E}[\tau_0]/T \rightarrow (1 - e^{-4c})/(4c)$, and the $\{|r| = 1\}$ occupations are $o(T)$; hence $\text{Corr} \rightarrow (1 - e^{-4c})/(4c)$.

One-sided observable. Each marginal $Z_i^u \sim N(0, 1)$ in Proposition 7.4, so $\mathbb{E}[Z_i^+] = 1/\sqrt{2\pi}$ and $\text{Var}(Z_i^+) = \frac{1}{2} - \frac{1}{2\pi}$ are u -independent, and (by uniform integrability — the fourth moments of $X_i/\sqrt{2T}$ are bounded — together with $\mathbb{P}(A_T^c) \rightarrow 0$ and Cauchy-Schwarz to remove the conditioning, Proposition 7.4) $\text{Corr}(X_1^+, X_2^+) \rightarrow (\mathbb{E}[g(U)] - \frac{1}{2\pi})/(\frac{1}{2} - \frac{1}{2\pi})$. By Price's theorem [16], $g'(u) = \partial_u \mathbb{E}[Z_1^+ Z_2^+] = \mathbb{P}(Z_1 > 0, Z_2 > 0)$, and Sheppard's orthant formula [19] gives $\mathbb{P}(Z_1 > 0, Z_2 > 0) = \frac{1}{4} + \frac{\arcsin u}{2\pi} = \frac{1}{2\pi}(\frac{\pi}{2} + \arcsin u)$, with $g(0) = \frac{1}{2\pi}$. Writing $g(U) = g(0) + \int_0^U g'$ and $\mathbb{P}(U > s) = e^{-4cs}$,

$$\mathbb{E}[g(U)] - \frac{1}{2\pi} = \int_0^1 g'(s) e^{-4cs} ds = \frac{1}{2\pi} \int_0^1 \left(\frac{\pi}{2} + \arcsin s \right) e^{-4cs} ds.$$

Integrating by parts and using $\int_0^1 e^{-4cs} (1 - s^2)^{-1/2} ds = \int_0^{\pi/2} e^{-4c \sin \theta} d\theta = \frac{\pi}{2} [I_0(4c) - L_0(4c)]$ (DLMF 11.5.2) yields the stated form. As $c \rightarrow \infty$, $I_0(z) - L_0(z) = \frac{2}{\pi z} + O(z^{-3})$ gives the $1/c$ tail; as $c \rightarrow 0$ both expressions tend to 1. $\square$

Remark 7.6. By Theorem 6.1 the limiting fields are decoupled, so the crossover is an initial-condition effect, not a transport coefficient: from a block of bound pairs $(Y_1(\cdot, 0) \equiv Y_2(\cdot, 0))$ the two decoupled flows decorrelate, the residual correlation measured by the $O(c)$ splits the run permits. In this picture $\rho(c)$ is the fading memory of a common origin — the two species released by a single split start out at the same place and are then carried apart by two independent flows, so the correlation surviving at time T records only how many such shared splits the run allowed. This is consistent with $C_{12} \equiv 0$ and $\sigma_{12} = 0$ (Proposition 3.3).

8 The Tracy–Widom regime

Now fix $q \in (0, 1)$ and study the integrated currents $N_1(s), N_2(s)$ across the origin from the step initial condition at a single time s , with standardised currents $\mathcal{O}^i = (N_i - \mathbb{E}N_i)/\sigma_i$, $\sigma_i \asymp s^{1/3}$. The marginal statements hold for *every* n by Proposition 3.2, via the time-changed generator $\tilde{\mathcal{L}}^{(n)} = \beta_n^{-1}\mathcal{L}^{(n)}$, under which the single-species rates are the n -free $q^{\mp 1}$. The joint statement (Theorem 8.4) uses the duality and is proved at $n = \infty$; it extends verbatim to $n = 2, 3$, where the duality is known [3], and to every n modulo the duality conjecture of [3, Rmk. 4].

8.1 Marginals

Theorem 8.1 (Tracy–Widom marginals). *For every $n \in \mathbb{N} \cup \{\infty\}$ and each species, $\mathcal{O}_s^i \Rightarrow F_2$, the GUE Tracy–Widom law.*

Proof. By Propositions 3.1(a) and 3.2, each species' current is an autonomous single-species ASEP — with rates $r_R = 1, r_L = q^2$ at $n = \infty$ and $q^{\mp 1}\beta_n$ at finite n , the factor β_n a pure time change removed by the standardisation. The step block (state 3 on $\{x \leq 0\}$, empty on $\{x > 0\}$) projects, per species, to the step initial condition, and step-ASEP currents are Tracy–Widom by [4, Thm. 1.4]. $\square$

Remark 8.2. Theorem 8.1 proves the Tracy–Widom conjecture of [3, §3.1.1] (stated there for $n = 2, 3$; here for every n), up to the standard equivalence between current and tagged-position asymptotics.

8.2 Triangular duality and q -Laplace decoupling

For the cross sector the dual has one walker of each species; let $\mathbf{r} = X_2 - X_1$ be the relative coordinate, which is driftless (the two species share the drift), finite-range, with a bounded perturbation at the origin from the binding. The triangular duality of Corollary 2.7 pairs finite dual configurations with the step sector, on which it takes a strikingly simple form.

Lemma 8.3 (step-sector duality identities). *Let η^0 be the step (state 3 on $x \leq 0$, empty on $x > 0$), let $X(\cdot)$ denote the free asymmetric walk (rates 1 right, q^2 left), and let (X_1, X_2) denote the two-particle dual — one particle of each species, each marginally a free walk by Proposition 3.1, jointly carrying the binding interaction on the contact set $\{|X_1 - X_2| \leq 1\}$. Then for all $a, b \in \mathbb{Z}$ and $s \geq 0$,*

$$\mathbb{E}_{\eta^0} \left[\eta_{i,a}(s) q^{2(a+N_{a+1}^+(\eta_i(s)))} \right] = \mathbb{P}_a(X(s) \leq 0), \quad (26)$$

$$\mathbb{E}_{\eta^0} \left[\eta_{1,a}(s) \eta_{2,b}(s) q^{2(a+N_{a+1}^+(\eta_1(s)))+2(b+N_{b+1}^+(\eta_2(s)))} \right] = \mathbb{P}_{(a,b)}(X_1(s) \leq 0, X_2(s) \leq 0), \quad (27)$$

and $\mathbb{P}_{(a,b)}(X_1(s) \leq 0) = \mathbb{P}_a(X(s) \leq 0)$ exactly.

Proof. Apply Corollary 2.7 with ξ a single species- i dual particle at a (respectively the pair at (a, b)). With one particle per species, $N^- \equiv 0$; and on the step, $N_{z+1}^+(\eta_i^0) = -z$ for $z \leq 0$, so every weight in $D^{\text{tri}}(\xi_s, \eta^0)$ equals $q^{2(X_i - X_i)} = 1$ and $D^{\text{tri}}(\xi_s, \eta^0) = \mathbf{1}\{X_i(s) \leq 0 \ \forall i\}$. On a finite lattice the displays are then exactly the duality relation; the extension to $\mathbb{Z}$ is the standard finite-propagation-speed approximation: couple the infinite system to its restriction to $[-K, K]$; for fixed s the two agree on $[-K/2, K/2]$ up to probability $O(e^{-cK})$, and (since the initial condition is the step) the event that any particle lies beyond $K/2$ at time s — the only way the right-tail counts N^+ in the observables can disagree — has super-exponentially small probability. We have additionally verified (26)–(27) exactly, by matrix exponentiation of the full forward dynamics on 8 sites against the one- and two-particle dual sectors. The marginal consistency is Proposition 3.1: the species-1 coordinate of the dual pair is an autonomous free walk. $\square$

The identities (26)–(27) show precisely what the two-particle dual computes: the currents enter through the q -deformed weights q^{2N^+} , not linearly. Summing the telescope

$$q^{2N_{a+1}^+(\eta_i)} - q^{2N_a^+(\eta_i)} = (1 - q^2) \eta_{i,a} q^{2N_{a+1}^+(\eta_i)} \quad (28)$$

over half-lines converts them into exact joint q -Laplace formulas, and the decoupling theorem of this regime is the following contact representation.

Theorem 8.4 (q -Laplace decoupling: exact contact representation). *At $n = \infty$, from the step initial condition, for all $s \geq 0$,*

$$\mathbb{E}[q^{2N_i(s)}] = 1 - (1 - q^2) \sum_{a \geq 1} q^{-2a} \mathbb{P}_a(X(s) \leq 0), \quad (29)$$

$$\text{Cov}(q^{2N_1(s)}, q^{2N_2(s)}) = (1 - q^2)^2 \sum_{a, b \geq 1} q^{-2(a+b)} C_s(a, b), \quad (30)$$

where $C_s(a, b) := \mathbb{P}_{(a,b)}(X_1(s) \leq 0, X_2(s) \leq 0) - \mathbb{P}_a(X(s) \leq 0) \mathbb{P}_b(X(s) \leq 0)$ is the connected pair kernel and all sums converge absolutely. The kernel is generated entirely by contact: $|C_s(a, b)| \leq \mathbb{P}_{(a,b)}(\tau_c \leq s)$, with τ_c the first meeting time of the two dual walkers, and $C_s \equiv 0$ if the binding interaction is removed. In particular the two currents decouple in the q -Laplace sense: $|\text{Cov}(q^{2N_1}, q^{2N_2})(s)| \leq \mathbb{E}[q^{2N_1(s)}] \rightarrow 0$.

Proof. (28) is $N_a^+ = N_{a+1}^+ + \eta_{i,a}$; summed over $a \geq m$ it telescopes to $1 - q^{2N_m^+(\eta_i)}$ (the configuration is right-finite, so $N_a^+ \rightarrow 0$), and $N_1^+(\eta_i) = N_i$. Taking expectations at $m = 1$ and inserting (26) weighted by q^{-2a} gives (29); the double telescope with (27) gives

$$\mathbb{E}\left[\frac{(1 - q^{2N_1})(1 - q^{2N_2})}{(1 - q^2)^2}(s)\right] = \sum_{a \geq 1} \sum_{b \geq 1} q^{-2(a+b)} \mathbb{P}_{(a,b)}(X_1(s) \leq 0, X_2(s) \leq 0),$$

and expanding the product and subtracting the marginal identities yields (30). Absolute convergence: to reach $\{\leq 0\}$ from a the walk makes at least a jumps, so $\mathbb{P}_a(X(s) \leq 0) \leq \mathbb{P}(\text{Poi}((1+q^2)s) \geq a)$ decays super-exponentially in a for fixed s , beating q^{-2a} ; this justifies Fubini throughout. Contact generation: realise the pair (X_1, X_2) and an independent pair $(\tilde{X}_1, \tilde{X}_2)$ on one probability space, with identical trajectories until the first time τ_c at which the walkers come within distance 1 — possible since off the contact set the pair dynamics consists of the same independent free jumps. Then $C_s(a, b) = \mathbb{E}[\mathbf{1}\{X_1(s) \leq 0, X_2(s) \leq 0\} - \mathbf{1}\{\tilde{X}_1(s) \leq 0, \tilde{X}_2(s) \leq 0\}]$, which vanishes on $\{\tau_c > s\}$; hence $|C_s(a, b)| \leq \mathbb{P}(\tau_c \leq s)$, and $C_s \equiv 0$ for independent walkers. Finally, for $U, V \in [0, 1]$

one has $|\text{Cov}(U, V)| \leq \min(\mathbb{E}U, \mathbb{E}V)$, and $\mathbb{E}[q^{2N_1(s)}] \rightarrow 0$ since $N_1(s) \rightarrow \infty$ in probability (the hydrodynamic limit of the autonomous ASEP marginal from step has strictly positive current) and $q^{2N_1} \leq 1$. $\square$

Conjecture 8.5 (linear covariance). *At every n , from the step initial condition, $\text{Cov}(N_1, N_2)(s) \sim c(q)\sqrt{s}$ as $s \rightarrow \infty$ with $c(q) \in (0, \infty)$; consequently $\text{Corr}(N_1, N_2) \asymp s^{-1/6} \rightarrow 0$ and the two Tracy–Widom currents are asymptotically uncorrelated. At $q = \frac{1}{2}$, Monte Carlo gives $c = 0.099 \pm 0.003$, with the $\sqrt{s}$ scaling holding over $s \in [50, 3.3 \times 10^4]$ and the correlation decaying at the $s^{-1/6}$ rate (§10).*

The linear covariance is not a finite-mass dual observable: by (26)–(27) the duality reaches the currents only through the weights q^{2N^+} , and extracting $\mathbb{E}[N_1 N_2]$ requires the full tower of joint q -moments (same-species dual particles in unbounded number) — the same difficulty as the joint law itself, left open in §11. This, and not a missing estimate, is why the covariance asymptotics are stated as a conjecture.

The natural candidate for the conjectured coefficient $c(q)$ is the contact combination $(1+q^4)\tau_0 - q^2(\tau_{+1} + \tau_{-1})$, $\tau_r = \int_0^s \mathbb{P}(\mathbf{r}(t) = r) dt$, since (exactly as in the proof of Theorem 7.5) the per-state increment second moments of $\mathbf{s} = X_1 + X_2$ and $\mathbf{r}$ coincide off contact ($a_r = c_r$ for $|r| \geq 2$) while at contact $a_0 - c_0 = 4(1+q^4)$, $a_{\pm 1} - c_{\pm 1} = -4q^2$. Two corrections to an earlier version of this draft are required here, and we record them precisely.

Proposition 8.6 (contact occupations of the relative walk). *The relative walk $\mathbf{r}$ is reversible with respect to the measure $m(0) = q^{-2}$, $m(r) = 1$ ($r \neq 0$): detailed balance across the bond $\{0, 1\}$ reads $m(0) \cdot q^2 \varepsilon = m(1) \cdot \varepsilon$ (split out of the bound state against merge into it), and is immediate at the swap and at all bulk bonds. Consequently*

$$\frac{\tau_0(s)}{\tau_{\pm 1}(s)} \longrightarrow q^{-2} \quad (s \rightarrow \infty),$$

the bound state being overweighted by exactly the stickiness ratio q^{-2} , and

$$(1+q^4)\tau_0 - q^2(\tau_{+1} + \tau_{-1}) \sim \frac{1-q^4}{q^2} \sqrt{\frac{s}{\pi(1+q^2)}}.$$

Proof. Reversibility: the bulk rates ($1+q^2$ each way) and the swap (q^2 each way between ± 1) are symmetric, so m is constant off the origin; the only nontrivial bond is $0 \leftrightarrow \pm 1$, with rate $q^2 \varepsilon$ out (split) and ε in (merge), giving $m(0) = q^{-2}m(\pm 1)$. The walk is recurrent (off a finite set it is the symmetric nearest-neighbour walk) and satisfies the hypotheses of Lemma 4.4 with $a = 1+q^2$, the swap being part of the finite defect, so $\tau_r(s) \sim m(r)\sqrt{s/(\pi(1+q^2))}$, and the two displays follow. Numerically (exact integration of the master equation at $q = \frac{1}{2}$): $p_t(0, 0)/p_t(0, 1) = 4.002$ at $t = 3000$ against $q^{-2} = 4$, and the combination divided by $\sqrt{s}$ is 1.875 at $s = 3000$ against the predicted limit 1.892, the deficit decaying as $s^{-1/2}$. $\square$

Remark 8.7 (the leading constant is open). An earlier version of this draft asserted the exact identity $\text{Cov}(N_1, N_2) = (1+q^4)\tau_0 - q^2(\tau_{+1} + \tau_{-1})$ together with the claim that the three occupations share a common leading coefficient, concluding $\text{Cov} \sim (1-q^2)^2 c' \sqrt{s}$. Both steps fail. First, the occupations do *not* share a coefficient: by Proposition 8.6 the combination carries the weight $q^{-2}(1-q^4)$, not $(1-q^2)^2$ — the bound state’s q^{-2} overweighting is precisely the residual trace of the binding interaction, and it changes the order of vanishing as $q \rightarrow 1$ from quadratic to linear in $1-q^2$. Second, the identity itself cannot hold with the occupations of a bound-started walk: forward Monte Carlo

of the full lattice dynamics at $q = \frac{1}{2}$ (4.5×10^4 trials, §10) gives $\text{Cov}(N_1, N_2) = (0.099 \pm 0.003)\sqrt{s}$, whereas $(1 + q^4)\tau_0 - q^2(\tau_{+1} + \tau_{-1}) \sim 1.89\sqrt{s}$ and the earlier claimed form evaluates to $0.28\sqrt{s}$. The structural reason is now transparent from Lemma 8.3: the two-particle dual computes the q -deformed currents (30), not the linear ones, so any occupation formula for $\text{Cov}(N_1, N_2)$ necessarily carries the nontrivial q^{2N^+} weighting. Identifying $c(q)$ in Conjecture 8.5 is open; the $\sqrt{s}$ scaling itself is confirmed numerically over $s \in [50, 3.3 \times 10^4]$.

Remark 8.8 (why there is no ballistic term). The step block is deterministic and species- i particles enter $\{x > 0\}$ only across the bond $(0, 1)$, so $\delta N_i = \delta J_i$ with J_i the single-bond integrated current; this localization gives the transport reading. Decomposing $J_i = M_i + A_i$ (martingale plus compensator), the martingale cross-bracket is exactly the contact carré du champ $\langle M_1, M_2 \rangle(s) = \int_0^s V_0(\eta_u) du$ with V_0 of (12) — only bound-pair hops and swaps move both species across one bond. The drift cross-terms $\text{Cov}(M_i, A_j), \text{Cov}(A_1, A_2)$ have a leading part equal to the equilibrium Green-Kubo cross-mobility $\sigma_{12} \cdot s$, which vanishes by Proposition 3.3. So $\text{Cov}(N_1, N_2)$ carries no $O(s)$ ballistic part, and the surviving $\sqrt{s}$ is the transient contact contribution evaluated above by duality. The forward decomposition thus explains the absence of an $O(s)$ term but does not *by itself* yield the constant: M_i and A_i each have $\Theta(s)$ variance and their cancellation to the $O(\sqrt{s})$ remainder is a transport-coefficient near cancellation, in contrast to the exact, term-by-term off-contact identity $a_r = c_r$ supplied by the duality.

Remark 8.9. Theorem 8.4 is the strongest joint statement we prove unconditionally: an exact identity, valid for all s , in which the inter-species coupling enters only through the contact kernel C_s — the q -Laplace analogue of uncorrelatedness. For the linear currents, Conjecture 8.5 is supported by the exact $\sqrt{s}$ scaling in the simulations and by the occupation analysis of Proposition 8.6; a proof would require joint control of the full q -moment tower, and even then says nothing about higher joint cumulants.

9 The asymmetry-exponent phase picture

The two regimes are endpoints of a one-parameter family of scalings, and the following observations — established at the level of the explicit rate algebra and extensive simulation, but not with the rigour of §§6–8 — place them in a single picture. Tune

$$q = 1 - cT^{-\alpha}, \quad \alpha \geq 0, \ c > 0, \quad (31)$$

and observe the integrated currents at time T ; the splits per particle are $\sim T(1 - q^2) \sim 2cT^{1-\alpha}$, governing the limiting cross-correlation.

- $\alpha < 1$: $\rightarrow \infty$ splits per particle, the species mix, $\text{Corr}(N_1, N_2) \rightarrow 0$; the marginals are Tracy-Widom (the asymmetry is still macroscopic), and $\alpha = 0$ is the fixed- q regime of §8.
- $\alpha = 1$ (critical): $O(c)$ splits, $\text{Corr} \rightarrow \rho(c) \in (0, 1)$, the crossover of Theorem 7.5; $\gamma \sim c/T$ vanishes, the KPZ nonlinearity switches off, and the limit is the Edwards-Wilkinson pair of §6.
- $\alpha > 1$: $\rightarrow 0$ splits, bound pairs survive, $\text{Corr} \rightarrow 1$ continuously — the locked limit of Remark 2.3.

A single axis thus interpolates from decoupled Tracy-Widom ($\alpha = 0$) through the EW crossover ($\alpha = 1$) to locking ($\alpha \rightarrow \infty$); the hydrodynamics decouple for every α, n (Proposition 3.1), only the

fluctuation-level coupling depending on the scaling. The crossover correlation $\rho(c)$ is not a transport coefficient: $C_{12} \equiv 0$ and (numerically) $\sigma_{12} \approx 0$ (measured -0.024 ± 0.022 at $c = 1$, against the front value $\rho(1) \approx 0.27$), so what remains is a transient, shared-source correlation — the species released by a common split carrying correlated currents, $\rho(c)$ the decaying memory of the perfectly correlated block initial condition. To our knowledge this combination (Edwards–Wilkinson class, diagonal compressibility, vanishing cross-mobility, nonzero initial-condition-driven cross-correlation with a $\rho(c) \sim 1/c$ crossover) matches no model in the multi-species fluctuating hydrodynamics or macroscopic fluctuation theory literature.

We record three further numerical observations, consistent across n (and into the negative- n window of Remark 2.3):

- *Crossover time.* At fixed q near 1 the Tracy–Widom statistics emerge only past $t_{\text{cross}} \sim (1 - q)^{-3/2}$; observed earlier the marginal looks Gaussian (the $\alpha \geq 1$ behaviour at finite time). At $q = \frac{1}{2}$ the law is clean by $T \sim 10^3$ (skewness $\rightarrow -0.21$, the F_2 value).
- *Subleading variance.* $\text{Var}(N_i) = A s^{2/3} + B s^{1/2} + \dots$, a relative-mode (Popkov–Schütz-type) correction; the standardised remainder $\sigma_{\text{extra}}^2 \asymp s^{-1/6}$ decays at the same rate as the cross correlation ($\sigma_{\text{extra}} \approx 0.39$ vs 0.36 at $n = 2$ vs 13 , $T = 3.3 \times 10^4$).
- *Multipoint uncorrelatedness.* The cross-species height functions are uncorrelated at all spatial separations ($n = 2$, $q = \frac{1}{2}$, $T = 3.3 \times 10^4$), so the decoupling extends from the integrated current to the spatial process, and it is n -independent at that level.

10 Numerical evidence

We summarise the Monte Carlo simulations that corroborate the theorems and the phase picture. Lattice simulations use the exact $n = \infty$ rate table (1) from a block initial condition of bound pairs; dual computations use the two- and few-particle generators directly. All runs store only currents and scalars (not lattice states), with 5×10^4 trials unless noted.

10.1 The Edwards–Wilkinson regime

The single-species current standard deviation fits the EW law $\text{std}(N_1) \sim T^{1/4}$ across $T \in [100, 10^4]$ (local slopes 0.22 – 0.28 , fit exponent 0.244), excluding the $T^{1/2}$ that a bound-pair third field would produce, consistent with the autonomous single-species drift of Proposition 6.11. The initial-condition crossover matches the closed forms of Theorem 7.5 to sampling error, with the predicted $1/c$ tail (Figure 1); the simpler full-position observable $(1 - e^{-4c})/(4c)$ matches equally well. Direct measurement gives $\text{Corr}(J_1, J_2) = -0.02 \pm 0.02$ (vanishing cross-mobility) and $C_{12} = 0$ within error, confirming that $\rho(c)$ is neither a static nor a dynamic transport coefficient.

10.2 The Tracy–Widom regime

The covariance scales as $\text{Cov}(N_1, N_2) \sim T^{1/2}$ (Conjecture 8.5), confirmed by exact augmented-matrix occupation-time computation ($\tau_0 \sim T^{0.505}$, $\text{Cov} \sim T^{0.504}$ at $q = \frac{1}{2}$) and by forward simulation (Figure 2, left). A dedicated forward run (4.5×10^4 trials, $T \leq 800$, $q = \frac{1}{2}$) pins the prefactor: $\text{Cov}(N_1, N_2) = (0.099 \pm 0.003)\sqrt{T}$, in agreement with the $n = \infty$ data of Figure 2 ($\approx 0.105\sqrt{T}$ over $T \in [10^2, 10^4]$); the same run reproduces the Borodin–Corwin–Sasamoto variance $2^{-8/3}(\gamma T)^{2/3} \text{Var}(\chi_2)$, $\gamma = 1 - q^2$, to within 10%, and exact integration of the relative-walk master equation gives $\tau_0/\tau_1 \rightarrow q^{-2}$ as in Proposition 8.6. Both theoretical candidate prefactors

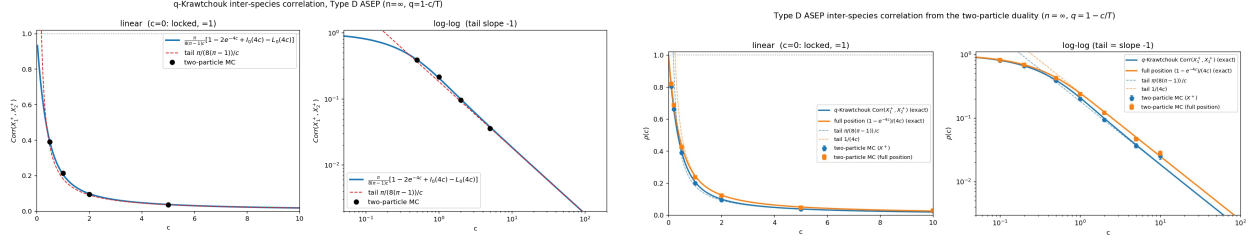


Figure 1: The initial-condition crossover of Theorem 7.5. Left: the one-sided observable $\text{Corr}(X_1^+, X_2^+)$ (Bessel-Struve closed form, curve) against two-particle Monte Carlo (dots), with the $\frac{\pi}{8(\pi-1)}c^{-1}$ tail. Right: the full-position observable $\text{Corr}(X_1, X_2) \rightarrow (1 - e^{-4c})/(4c)$ against Monte Carlo. Both run from perfectly correlated data ($\text{Corr} \rightarrow 1$ as $c \rightarrow 0$) to the decoupled $1/c$ tail.

of Remark 8.7 are excluded, so the constant carries a nontrivial front factor. The marginals converge to F_2 (Figure 2, middle: CDF/histogram; skewness $\rightarrow -0.21$, excess kurtosis $\rightarrow 0.09$, the GUE Tracy-Widom values), and the limit is n -independent (Figure 2, right: $n = 2$ and $n = 13$ collapse onto F_2). The standardised correlation accordingly decays, $\text{Corr}(N_1, N_2) \sim s^{-1/6} \rightarrow 0$, the asymptotic decoupling of Conjecture 8.5.

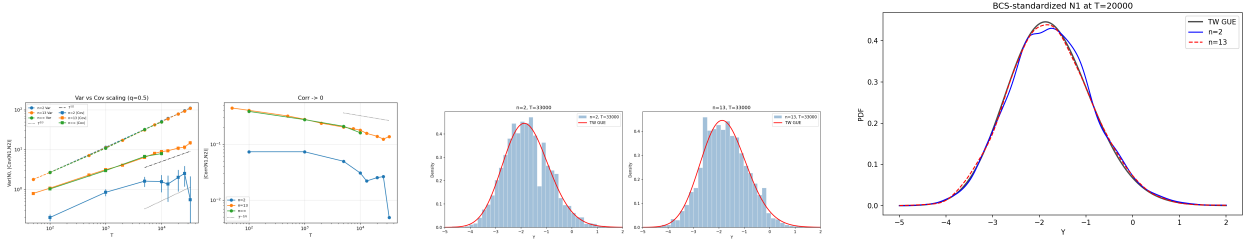


Figure 2: The Tracy-Widom regime. Left: $\text{Cov}(N_1, N_2) \sim T^{1/2}$, the uncorrelatedness of Conjecture 8.5. Middle: the marginal current against the GUE Tracy-Widom law F_2 at $T = 3.3 \times 10^4$. Right: n -independence — the standardised marginals at $n = 2$ and $n = 13$ collapse onto F_2 .

11 Discussion

The type D bound pair is a slow, near-conserved internal mode, yet in both scaling regimes the species decouple in the limit, for one structural reason (Remark 3.4): the asymmetry lives only in the marginals (Proposition 3.1), and the coupling is invisible to both equilibrium transport coefficients (Proposition 3.3). What is left of the coupling is a transient, initial-condition-driven object: in regime (A) the entire crossover $\rho(c)$ (Theorem 7.5); in regime (B) the contact kernel of the exact q -Laplace representation (Theorem 8.4) and, conjecturally, the sub-leading $s^{-1/6}$ correlation that decays to leave the two currents uncorrelated (Conjecture 8.5). The two regimes are governed by one object — the contact occupations $(1 + q^4)\tau_0 - q^2\tau_{\pm 1}$ of the relative walk — in two windows: at $\alpha = 1$ the split rate is $O(\varepsilon) \rightarrow 0$, $\tau_0 = \Theta(T)$, the crossover $O(1)$; at $\alpha = 0$ the split rate is $\Theta(1)$, $\tau_0 = \Theta(\sqrt{T})$, the correlation sub-leading.

We emphasise the scope: both results are statements of *decoupling*. In regime (A) the Gaussian limiting fields are uncoupled in drift and noise; in regime (B) the two currents decouple exactly at the level of the joint q -Laplace observable, and — conjecturally, with exact numerical scaling — are

asymptotically uncorrelated. We do not address the finer question of whether the regime (B) limit is a *product* (joint) law; the higher-order joint structure of the two currents, the linear covariance asymptotics of Conjecture 8.5, and its constant $c(q)$ are facets of one open problem: the joint q -moment tower beyond two dual particles.

The Edwards–Wilkinson decoupling holds at *arbitrary* densities (ρ_1, ρ_2) , not only on the symmetric line (Theorem 6.1): every ingredient is density-free — the current decoupling (Proposition 3.1), the cross-noise cancellation $\mathbb{E}_\nu[V_x] = 0$ (Proposition 3.3), the orthogonality $V_x \perp$ density (Lemma 6.3), and the kernel bound (Theorem 5.2), while the diagonal noise is per species — so the two fields decouple into linear stochastic heat equations with species-dependent noise intensities $\sqrt{2\chi_i D}$, $\chi_i = \rho_i(1 - \rho_i)$, and the bound-pair mode never produces a third hydrodynamic field (Corollary 6.9, Remark 6.10). The one genuinely open direction is the non-stationary EW analysis from block initial data, for which $\rho(c)$ is the exact limiting cross-correlation and Proposition 7.4 the natural tool.

Status. The $n = \infty$ self-duality is Theorem 2.6(ii) (two-site interlacing verified by computer algebra, extended to all L by the induction of [3]). The computer-algebra and numerical verifications cited in Theorem 2.6, Proposition 3.2, Remark 2.9, Lemmas 6.3, 6.6 and 8.3, and §10 are reproduced by scripts to be included as ancillary files. In regime (A): Propositions 3.3, 6.11, Corollary 6.9, Lemmas 6.3, 6.4, 4.1–4.4, 5.1, 7.1, 7.2, Theorems 5.2, 7.5 and Proposition 7.4 are complete and self-contained; Proposition 6.7 rests on the exact, machine-verified orthogonality relation (10) and is complete modulo Lemma 6.6 (the dressed-mass estimate, perturbative and numerically supported) and the infinite-volume step recorded in Remark 2.9; Theorem 6.1 is complete modulo the classical single-species WASEP fluctuation result (used for the drift via Proposition 3.1 and the diagonal noise) and the classical tightness/uniqueness [13]. In regime (B): Theorem 8.1 (marginals, every n , via Proposition 3.2) is complete; Theorem 8.4 (q -Laplace decoupling) is complete at $n = \infty$ — the triangular reduction is Lemma 8.3 — extends to $n = 2, 3$ via [3], and for other finite n rests on the duality conjecture of [3, Rmk. 4]; the linear-current statement is Conjecture 8.5, its constant open (Remark 8.7). Section 9 is numerical/exploratory and frames the rigorous results within the (α, q, n) picture.

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